

BLOCKFELL

How to Play

A player's guide to mining, crafting, building and surviving the night — from your first tree and first harvest to a cosy cottage, a busy village, the fiery Cinderdeep and the floating islands of the Starhollow.



Original game, art and sound. Not affiliated with Mojang or Microsoft.

Welcome

Blockfell is an open-world building game made of blocks. You explore a world that is generated as you move, dig up materials, craft them into tools and buildings, and look after your health and hunger — especially at night, when hostile creatures come out.

Quick start

1. Double-click **Blockfell.html**. It opens in your web browser (Chrome, Edge or Firefox).
2. Click **Singleplayer**, then **Create New World**, then **Create New World** again at the bottom of the screen.
3. When the world appears, **click once** on the game to capture the mouse. Move with W A S D and look around with the mouse.
4. Walk up to a tree, point the white cross at the trunk and **hold the left mouse button** until the block breaks. Walk over it to pick it up.
5. Press E to open your inventory and turn the log into planks (section 5).
6. Press Esc at any time to pause. **Save and Quit to Title** saves your world.

What's new

2.13: Axolotls, Squid and Guardians (38). **2.12:** richer pixel art for every creature and person (shading, ears, sleeves, cuffs and shoes). **2.11.1:** name yourself with a Name Tag, and a third-person view. **2.11:** Treasure Maps and buried treasure, rainbows and Lightning Rods, Note Blocks, the Tune Box and Jukebox, and the Wandering Trader (37). **2.10:** Moth Wings, Fireworks, five new fish and Fish Jars, Spring Pads and the Hootlet (36). **2.9.1:** the Potion of Featherfall. **2.9:** Potatoes, Milk, the Powderbug, the Gloommoth and Sporecap colours (35). **2.8.1:** the Spore Compass, and a longer Explorer's Guide. **2.8:** Spore Isle, the Sporecap, Spore Stew, the Spore Lamp and the Trident (34). **2.7:** saddles, riding, horses and the Lavastrider (33). **2.6:** spears, the crossbow, a charge bar and Flame Arrows (32). **2.5.1:** the version and build code shown on the title and pause screens. **2.5:** the Brewing Stand, six potions and six splash potions (31). **2.4.1:** a Hollowdrake Trophy for every win; replay the credits from the pause menu (23). **2.4:** tracks, Minecarts, Party Bags, cake and party things (25). **2.3:** snow day, Tim the fox cub, Name Tags, baby animals, dyed sheep (24). **2.2:** the Starhollow, the Hollowdrake, Starwings and the End (23). **2.1:** armour stands, dyed armour (19); Ashboars (22). **2.0:** the Cinderdeep, a world below (22); 2.0.1: Glimmerstone.

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1 Starting a game

Opening Blockfell

Blockfell is a single file, **Blockfell.html**. Double-click it and it opens in your web browser; nothing needs to be installed and no internet connection is needed. Use a recent desktop version of **Chrome, Edge or Firefox** with a keyboard and mouse.

> **Where your worlds are saved**
>
> Worlds are saved inside the browser you play in, not in the Blockfell.html file. To keep your worlds, keep using the same browser and the same copy of Blockfell.html, and don't clear that browser's site data. A world you start in Chrome won't appear in Firefox. To keep safe copies in Dropbox, use **Export World** and automatic backups (section 29). To play the same worlds on your Mac and iPad, open Blockfell from its web address and use **Dropbox sync** (section 27).

The title screen



The title screen, over a live view of a generated world. **Continue** takes you back into the world you played last.

BUTTON	WHAT IT DOES
Continue	Straight back into the world you played last (its name is on the button). Appears once you have a world.
Singleplayer	Opens your list of worlds.
Options…	Graphics, sound and mouse settings (section 28).
Quit Game	Leaves the game; you can then close the browser tab.

Choosing a world

Singleplayer shows your saved worlds, the one played last at the top and already selected, with the date last played and the game mode. Click a world to select it, then:

Play Selected World	Loads the world where you left it (double-clicking a world does the same).
Create New World	Opens the new-world screen described below.
Edit	Rename the world or make a backup copy of it.
Delete	Deletes the world for good, after asking you to confirm.
Re-Create	Starts a fresh world with the same seed and settings as the selected one.
Export World	Saves a backup file of the selected world into your backup folder, e.g. Dropbox › MineCraft (section 29).
Import World…	Restores a world from a backup file.
Dropbox…	Keeps your worlds in Dropbox to play them on your other devices (section 27).

Creating a new world



The Create New World screen.

World Name	Any name you like.
Game Mode	**Survival**: gather everything yourself and look after your health and hunger. **Creative**: unlimited blocks, flying, instant breaking.
Difficulty	See the table below. You can change it later from Options.
Seed	Leave blank for a random world. Typing the same seed always makes the same landscape, so you can share good worlds.
Structures	Adds **villages** (section 16), hidden **dungeons** (section 21) and small abandoned cobblestone ruins, each ruin with a chest of random loot (coal, iron, torches, food, arrows and sometimes tools or armour).
Bonus Chest	Places a chest next to where you start, marked by a torch. It holds 4 logs, 8 planks, a wooden pickaxe, a wooden axe, 3 apples and 4 sticks — a good choice for a first game.
Game Rules…	**Keep inventory after death** (off by default), **Advance time of day**, **Spawn creatures** and **Natural health regeneration** (all on by default).

Difficulty levels

DIFFICULTY	HOSTILE CREATURES	HUNGER AND HEALTH
Peaceful	None appear.	Health and hunger refill on their own.
Easy	Up to 5 nearby; they do less damage.	An empty hunger bar lowers health, but never below 5 hearts.
Normal	Up to 8 nearby; normal damage.	Starving can take you down to half a heart.
Hard	Up to 12 nearby; 50% more damage and faster archers.	Starving can kill you.

Click **Create New World**. A loading screen shows the land being generated around you; after a few seconds you are placed on solid ground in daylight. When you come back to a saved world, you continue exactly where you stopped.

> **Picking up where you left off**
>
> If Blockfell is closed while you are playing — the tab or browser closed, the iPad app swiped away — it opens straight back into that world next time, where you were (Blockfell saves whenever it goes into the background, and every minute anyway). After **Save and Quit** it starts at the title screen instead, with **Continue**. Turn this off with **Reopen Last World** in Options.

2 Controls

Click once on the game to capture the mouse pointer, so that moving the mouse turns your view. Press Esc to get the pointer back; this also opens the pause menu.

Moving

W A S D	Walk forward, left, back, right
Mouse	Look around
Space	Jump; swim upwards in water (even up a waterfall)
Left Ctrl	Sprint while moving forward (or double-tap W)
Left Shift	Sneak: slow, and you won't walk off edges
Space ×2	Creative only: double-tap to start or stop flying
Space / Shift	While flying: go up / go down

Screens

E	Open or close your inventory
Esc	Pause menu; closes any open screen
M	Sound on or off
F1	Hide the on-screen display (for screenshots)
F3	Show position, frame rate and other details

Doing things

Left mouse (hold)	Break the block you're pointing at; attack a creature
Right mouse	Place a block; open a crafting table, furnace, chest or Rune Table; open or close a door; sleep in a bed; fill or empty a bucket; put on armour; use a spawn egg; trade with a villager; till soil with a hoe; plant seeds; use bone meal; feed a Fellhound, or tell yours to sit or follow; open a gate or trapdoor; write on a sign; plant a flower in a pot; hang or change a painting; dress an armour stand
Right mouse (hold)	Eat the food in your hand; draw a bow, then release to shoot
Middle mouse	Pick block: select the block you're pointing at from your hotbar
1–9, wheel	Choose a hotbar slot
Q	Drop one of the held item (Ctrl + Q drops the stack)
F	Swap the held item with your off-hand slot

Inside inventory screens

Left-click	Pick up, put down or swap a stack
Right-click	Pick up half a stack, or put down one item
Drag	Left-drag spreads a stack evenly; right-drag puts one in each slot
Shift + click	Move a stack to the other part of the screen; on a crafting result, craft as many as you can
Double-click	Gather all items of that kind onto the pointer
1–9 over a slot	Swap it with that hotbar slot
Click outside	Drop what you're holding into the world

Playing on iPhone and iPad

On a phone or tablet Blockfell switches to on-screen controls by itself. Hold the phone sideways. Your left thumb walks, your right thumb looks around and does things.



1. **Joystick.** Touch anywhere in the lower left and the stick appears under your thumb. Push past the rim to **sprint**.
2. **Drag** anywhere else to look around. **Tap** to place or use: doors, chests, gates, beds, villagers. Tap a creature to **attack** it.
3. **Touch and hold** to mine at the cross; the ring shows the hold has begun. Hold with food to eat, or with a bow to draw it.
4. **Jump.** Hold to swim up. In Creative, double-tap to fly.
5. **Sneak** on and off (lit while on). While flying it takes you down.
6. **Inventory** (the catalogue in Creative). The × at the top right closes it.
7. **Pause**: the game menu, with Options and Save and Quit.
8. **Hotbar.** Tap a slot to choose it; hold it to drop one.

Getting Blockfell onto the device

- An iPhone or iPad (iOS 15 or later) can't run **Blockfell.html** sent by WhatsApp or mail, or kept in Files. Open it from its **web address** in Safari instead: **avizent.github.io/blockfell** (or the Blockfell page on claude.ai).
- In Safari tap **Share › Add to Home Screen**: Blockfell opens full screen with its own icon, safe from Safari's clear-out of sites not visited lately.
- **No internet needed** once it has been opened from its web address. When a newer version is online, the title screen offers **Restart** (in a world: menu › **Save and Update**); your worlds are kept.

Settings

- **Touch Controls** in Options: **Auto** (on for phones and tablets), **On** or **Off**.
- Phones start with Render Distance 5 and Render Scale 75%. Lower them if the game feels jerky.

In inventories

- **Tap** a stack to pick it up, tap a slot to put it down; or **drag** it there.
- **Hold** a stack to take half. Carrying one, hold a slot to put down one (slide on: one per slot).
- **Drag** a stack across slots to share it out. **Swipe** the catalogue and trade list.

Moving worlds to and from the iPad

- **Dropbox sync** (section 27) keeps the same worlds on both, up to date.
- By hand: **Export World** on the computer; on the iPad **Import World… › Import From File… › Dropbox › MineCraft**.
- Back again: Export World on the iPad saves to **Files › Downloads**; move it into Dropbox › MineCraft and import it.
- Export and import don't work inside the claude.ai page.

3 Reading the screen



Playing in Survival. The white cross in the middle is your aim: the block or creature it points at is what you break, hit or use (up to about 4½ blocks away).



1. **Armour.** Shows only when you wear armour. Each full icon is 2 armour points; more armour means less damage taken.
2. **Health.** 10 hearts (20 points). At zero you die and respawn.
3. **Experience bar.** Fills towards the next level.
4. **Hotbar.** Your 9 quick slots; the white frame is the selected slot. Numbers show how many you have.
5. **Experience level.** Earned from mining ore, smelting and defeating creatures.
6. **Hunger.** 10 drumsticks. Eat to keep it high; a full bar heals you.
7. **Your hand** and the item you're holding.

· Air bubbles appear above the hunger bar when you are under water. Messages and new advancements pop up at the edges of the screen.

4 Your first day, step by step

In Survival you start with nothing. The aim of the first day is simple: get wood, make tools, find coal for torches and have somewhere safe to be when the sun goes down (after about 10 minutes).

1. **Collect wood.** Point at a tree trunk and hold the left mouse button. Breaking by hand is slow but works for wood. The block drops as a small item; walk over it to pick it up. Collect 5–10 **logs**.
2. **Make planks.** Press E. Your inventory has a small 2×2 crafting grid. Click a stack of logs to pick it up, right-click a grid square to put one log down, and 4 **planks** appear on the right. Click the result to take it, or Shift-click it to use up all your logs at once.
3. **Make a crafting table.** Fill all four squares of the 2×2 grid with planks. Put the **Crafting Table** in your hotbar, close the inventory, look at the ground and right-click to place it. Right-click the table to open the larger 3×3 grid.
4. **Make sticks and a wooden pickaxe.** Two planks stacked vertically make 4 **sticks**. Then, on the crafting table, 3 planks across the top and 2 sticks down the middle make a **wooden pickaxe**. A wooden sword and axe are useful too.
5. **Mine stone.** Find grey stone on a hillside or at the bottom of a hole and mine it with your pickaxe; it drops **Cobblestone**. Stone mined by hand drops nothing. Use cobblestone to make **stone tools**, which are twice as fast and last longer.
6. **Find coal and make torches.** **Coal ore** is stone with black specks, common on cliffs and in caves. Mine it for **Coal**. One coal above one stick makes 4 **torches**. No coal? Smelt a log in a furnace and it becomes coal.
7. **Build a furnace.** 8 cobblestone in a ring on the crafting table make a **Furnace**. Use it to cook meat, smelt iron and turn sand into glass (section 6).
8. **Get ready for night.** When the sky turns orange, build a small shelter from any blocks, or dig into a hillside and close the entrance behind you. Place torches inside: hostile creatures only appear in the dark. Once you have 3 wool (from sheep) you can craft a **bed** and sleep through the night (section 12).
9. **Find food.** Your hunger bar slowly empties as you move, jump and fight. Pigs, cows, sheep and chickens drop raw meat; cook it in the furnace for far more nourishment. Oak leaves sometimes drop apples.
10. **Go for iron.** **Iron ore** (stone with tan specks) is found underground below the height of the sea. It needs a **stone pickaxe or better**. It drops **raw iron**; smelt it into **iron ingots** for iron tools and armour.

Good habits

- Keep a sword, a pickaxe, food and torches in your hotbar.
- Don’t dig straight down: you may drop into a cave. Dig a staircase instead.
- Hold Shift near cliff edges and when building high; sneaking stops you walking off.
- Put a chest in your base and store what you don’t need to carry.

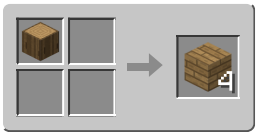
5 Crafting recipes

Your inventory has a **2×2** crafting grid; a **crafting table** gives a **3×3** grid. Put the ingredients in the pattern shown and the result appears on the right. Patterns can sit anywhere in the grid and can be mirrored left-to-right. Recipes marked INVENTORY OR TABLE fit in either grid.



A crafting table set up to make a stone pickaxe.

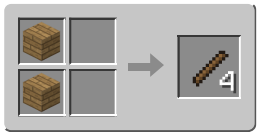
Basics



Planks ×4

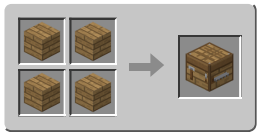
INVENTORY OR TABLE

Put a log in any slot.



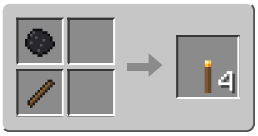
Stick ×4

INVENTORY OR TABLE



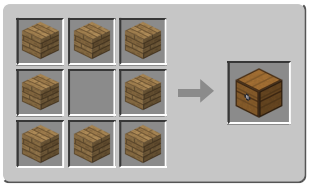
Crafting Table

INVENTORY OR TABLE



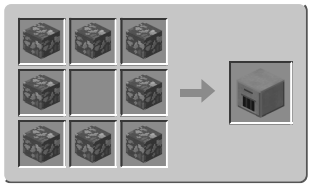
Torch ×4

INVENTORY OR TABLE



Chest

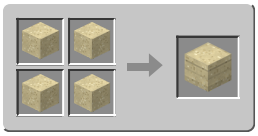
CRAFTING TABLE



Furnace

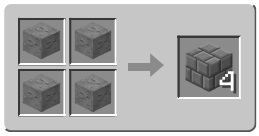
CRAFTING TABLE

Building blocks



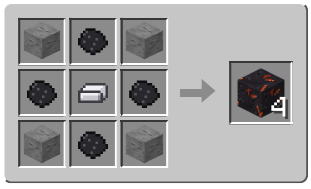
Sandstone

INVENTORY OR TABLE



Stone Bricks ×4

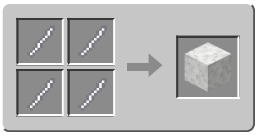
INVENTORY OR TABLE



Cinderstone ×4

CRAFTING TABLE

No lava needed (1.10). 4 make 4 Cinderstone Bricks.



White Wool

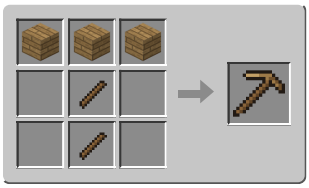
INVENTORY OR TABLE

Glass and smooth stone are made in the furnace, not on the crafting table (section 6). Birch and spruce logs make planks too. Recipes for beds, doors, slabs and stairs are in section 12, the bucket in section 13 and the Rune Table in section 14.

Tools and weapons

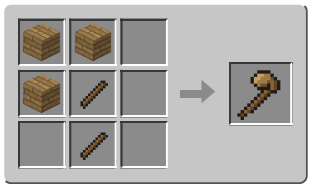
Each tool is made from sticks plus one material. The material decides the tool's tier:

Oak Planks → Wooden Cobblestone → Stone Iron Ingot → Iron



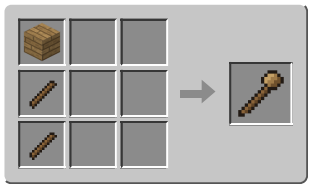
Pickaxe

CRAFTING TABLE



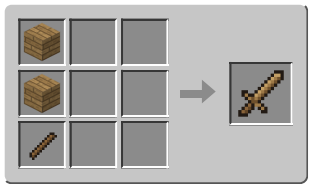
Axe

CRAFTING TABLE



Shovel

CRAFTING TABLE

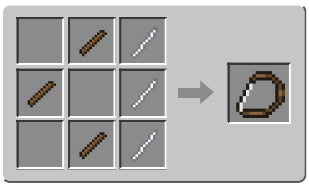


Sword

CRAFTING TABLE

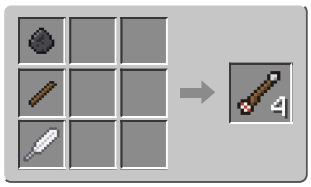
Ranged weapons

String comes from Bone Archers (and ruin chests), flint from breaking gravel (1 time in 10), feathers from chickens.



Bow

CRAFTING TABLE

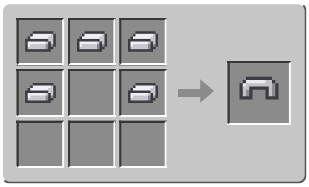


Arrow ×4

CRAFTING TABLE

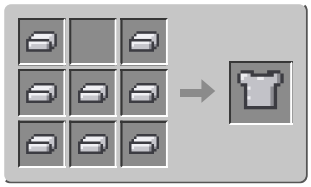
Armour

Shown with iron ingots. **Leather** (dropped by cows) makes a lighter set with the same patterns. Right-click a piece of armour in your hand to put it on, or place it in the armour slots of your inventory.



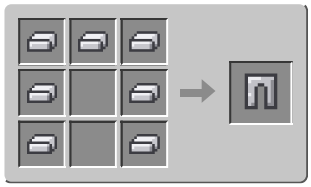
Helmet

CRAFTING TABLE



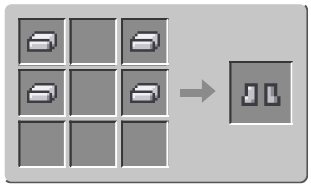
Chestplate

CRAFTING TABLE



Leggings

CRAFTING TABLE

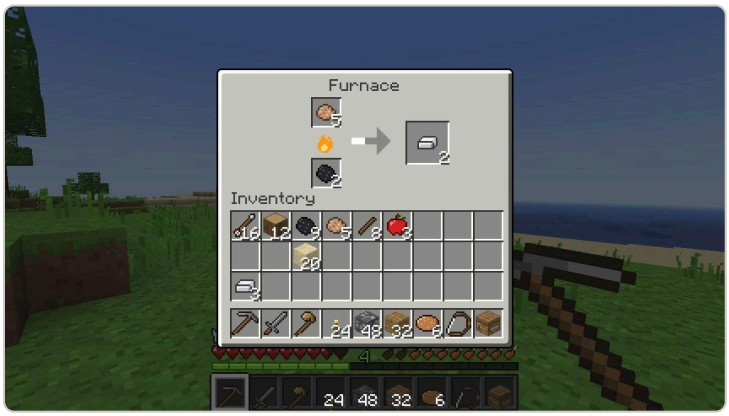


Boots

CRAFTING TABLE

ARMOUR POINTS	HELMET	CHESTPLATE	LEGGINGS	BOOTS	FULL SET
Leather	1	3	2	1	**7**
Iron	2	6	5	2	**15**

6 Furnaces, smelting and fuel



Raw iron (top) being smelted with coal (bottom). The flame shows fuel burning; the arrow shows progress.

Place a furnace and right-click it. Put the item to cook in the **top** slot and fuel in the **bottom** slot. Each item takes **10 seconds**. The finished items collect on the right; the furnace keeps working when you close the screen or walk away, as long as it is within your render distance.

Taking the results out gives you the experience earned by smelting. A working furnace glows and gives off light.

What to smelt

PUT IN	YOU GET
Raw Iron or **Iron Ore**	**Iron Ingot**
Sand or **Red Sand**	**Glass**
Cobblestone	**Stone**
Any log or **Coal Ore**	**Coal**
Raw Porkchop	**Cooked Porkchop**
Raw Beef	**Steak**
Raw Chicken	**Cooked Chicken**
Raw Mutton	**Cooked Mutton**
Raw Rabbit	**Cooked Rabbit**
Raw Trout or **Raw Perch**	**Cooked Trout** or **Cooked Perch**

Fuel

FUEL	ITEMS SMELTED
Lava Bucket (gives the bucket back)	**100**
Ember (from the Cinderdeep)	**24**
Coal	**8**
Any log, **Planks**, **Oak Stairs**	1½
Crafting Table, **Chest**, **Bow**	1½
Wooden tools, **Oak Door**	1
Oak Slab	¾
Stick	½

> **Tip:** coal is by far the best fuel. Early on, smelt one log into coal and use that coal to smelt eight more things.

7 Mining and tools

Anything can be broken with your bare hands, but the right tool is much faster, and some blocks drop nothing unless you use a pickaxe.

BLOCK	BEST TOOL	NEEDS	DROPS
Grass, dirt, sand, red sand, gravel, snow	Shovel	—	Itself (grass drops dirt; gravel sometimes flint)
Logs, planks, crafting table, chest	Axe	—	Itself
Stone, cobblestone, sandstone, bricks, stone bricks, terracotta, furnace, Rune Table	Pickaxe	Any pickaxe	Itself (stone drops cobblestone)
Coal ore	Pickaxe	Any pickaxe	Coal + a little experience
Iron ore	Pickaxe	**Stone pickaxe or better**	Raw iron
Rune ore (glows faintly)	Pickaxe	**Stone pickaxe or better**	1–3 rune shards + experience
Cinderstone and its bricks	Pickaxe	**Iron pickaxe**	Itself, after about 9 seconds
Monster Cage	Pickaxe	Any pickaxe	Experience only
Cactus, doors, beds	Any (axe for doors)	—	Itself; a door or bed drops once
Leaves	Sword or hand	—	Occasionally an apple or sticks
Glass	Any	—	Nothing: glass breaks
Bedrock (bottom of the world)	Cannot be broken.		

Tool tiers

TIER	SPEED WITH THE RIGHT TOOL	USES BEFORE BREAKING	TIME TO MINE STONE	SWORD DAMAGE
Hand	1×	—	7.5 s (drops nothing)	½ heart
Wooden	2×	59	1.1 s	2 hearts
Stone	4×	131	0.6 s	2½ hearts
Iron	6×	250	0.4 s	3 hearts

- A bar under a tool's icon shows wear, from green to red; when it runs out the tool breaks.
- Mining is 5 times slower while you are under water or in the air (jumping or falling), so stand on solid ground.
- Ores: coal from the bottom of the world up into the hills, iron up to about sea level, rune ore only below height 30; Ember and Fire Opal Ore in the Cinderdeep (22).

Placing blocks

Select a block in the hotbar, point at the face of a block and right-click: the new block goes against that face (never where you or a creature stands). Torches stand on a block or hang on a wall; slabs, stairs, doors and beds have their own rules (section 12). The off-hand slot places blocks too.

8 Health, hunger and experience

Health

You have 10 hearts. You lose health from creature attacks, falls, drowning and starving.

- **Falling:** the first 3 blocks are free; each block after that costs half a heart. Landing in water is safe.
- **Cactus:** touching one pricks you (half a heart each half second).
- **Drowning:** under water you have 10 air bubbles (about 15 seconds). When they run out you take damage until you surface.
- **Healing:** with 9 or more drumsticks of hunger you slowly regain health (half a heart every 4 seconds), which uses up food.

Hunger

The hunger bar empties slowly as you walk, sprint, jump, swim, fight and heal. Food also gives hidden "saturation", which is used up before the bar starts to drop, so cooked food keeps you going much longer.

- You can't sprint when hunger is 3 drumsticks or less.
- With an empty bar you slowly lose health (how far depends on difficulty).
- To eat, hold the food and keep the right mouse button pressed for about 1.5 seconds. You can only eat when not full.

Food

FOOD	DRUMSTICKS	NOTES
Cooked Porkchop	4	Best
Steak	4	Best
Cooked Mutton	3	
Cooked Chicken	3	
Cooked Perch	3	Long-lasting
Cooked Rabbit	2½	
Cooked Trout	2½	
Apple	2	From leaves
Spoiled Flesh	2	Barely filling
Raw Porkchop	1½	Cook it!
Raw Beef	1½	Cook it!
Raw Rabbit	1½	Cook it!
Raw Chicken	1	Cook it!
Raw Mutton	1	Cook it!
Raw Trout, **Raw Perch**	1	Cook it!

Death and respawning

If your health reaches zero, the **You Died!** screen appears. Choose **Respawn** to return to your bed (if you have slept in one) or the world's spawn point, with full health and hunger. Unless the *Keep inventory* game rule is on, everything you carried is dropped where you died — go back quickly to collect it.

Experience

Green orbs come from defeating creatures, mining ore and taking smelted items out of a furnace. Walk over them to collect them. Your level is shown above the hotbar and counts towards your score on the death screen.

9 Creatures and combat



Left to right: pig, cow, Shambler, chicken, sheep and Bone Archer.



New in 1.1, left to right: goat, rabbit, Crawler and Dustwalker.

Animals (harmless)

ANIMAL	HEARTS	WHERE	DROPS
Pig	5	Grassland and forests	1–3 raw porkchop
Cow	5	Grassland and forests	1–3 raw beef, 0–2 leather
Sheep	4	Grassland, forests, hills (24)	1 wool (its colour), 1–2 raw mutton
Chicken	2	Grassland and forests	1 raw chicken, 0–2 feathers
Goat	5	Mountains, snowy peaks, taiga.	1–2 raw mutton, 0–1 leather
Rabbit	1½	Deserts, taiga, some grass. Hops.	0–1 raw rabbit, 0–1 leather
Fellhound	4	Forests, taiga and snow, in packs. Attack only if you hit one. Tame with raw meat (18).	Nothing
Ashboar	10	Cinderdeep floors, in herds. Charge only if you hurt one (22).	1–3 raw Ashboar, 0–2 leather
Drifter	4	Starhollow islands. Floats away when hit (23).	1–2 Drift Silk
Tim (fox cub)	5	Taiga, snowy lands. Shy; tame with fish (24).	Nothing
John (snowman)	4	Built from Snow Blocks and Coal (24).	2–6 snowballs

Animals wander in daylight, avoid big drops, and run away when hit.

Hostile creatures

CREATURE	HEARTS	BEHAVIOUR AND DROPS
Shambler	10	Walks straight at you and hits. Burns in sunlight. Drops spoiled flesh, rarely an iron ingot.
Bone Archer	10	Keeps its distance and shoots arrows. Burns in sunlight. Drops bones, arrows and string.
Crawler	8	A wide, eight-legged creature that **climbs walls**. In daylight or bright light it leaves you alone unless you hit it. Too wide for one-block gaps. Doesn't burn. Drops 0–2 string.
Dustwalker	10	Takes the Shambler's place in deserts and badlands. **Doesn't burn in sunlight**, and its hits make you hungry. Drops spoiled flesh.

Hostile creatures appear in darkness (light level below 7), at night or in caves, usually at least 24 blocks from you. Thunderstorms are dark enough for them to appear in the open by day too.

Fighting

- Point at a creature and click the left mouse button. A sword does the most damage; axes are next best.
- **Critical hits:** hit while falling after a jump for 50% extra damage (you'll see sparks).
- Hits knock creatures back. Armour reduces the damage you take.
- **Bow:** needs arrows in your inventory. Hold the right mouse button to draw (longer = stronger), release to fire. Arrows that miss can be picked up again.
- Runes make weapons and armour stronger (section 14).

Staying safe at night

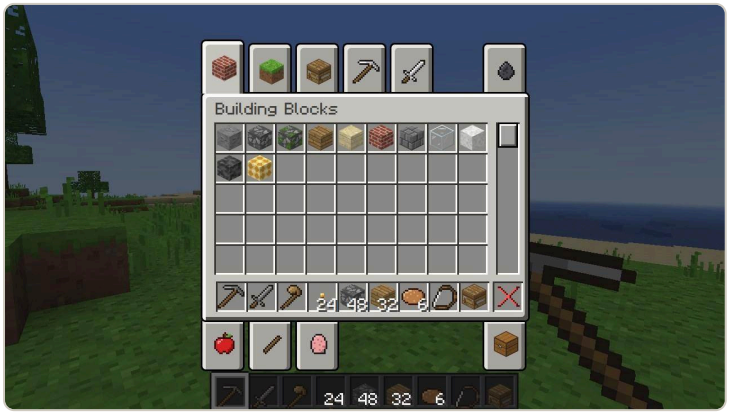
- Light your base: torches keep an area bright enough that nothing can appear there. Place them no more than about 12 blocks apart.
- Close up gaps in walls and roofs, and fit a door — hostile creatures can't open doors or break blocks.
- Crawlers climb, so give outdoor walls an overhang, or a roof.
- Fight Bone Archers from behind cover, or close the gap quickly.
- Sleep in a bed to skip the night — or survive until sunrise to earn **First Night**.
- A tamed Fellhound at your side fights whatever attacks you (section 18).



At night a Shambler (left) and a Bone Archer (right) approach, between two torches.

10 Creative mode

Creative is for building without limits. You have no health or hunger bar, blocks break instantly, blocks you place are never used up, and you can fly.



The Creative catalogue, open on the Building Blocks tab. The red cross at the bottom right is the destroy slot.

The catalogue

Press E to open it. Tabs along the top and bottom group every item: Building Blocks, Decoration Blocks, Natural Blocks, Functional Blocks, Tools, Combat, Food, Ingredients and Spawn Items, plus **Search** (type part of a name) and **Survival Inventory** (your armour, off-hand and full inventory).

- Click an item to pick up a full stack; right-click for one. Click a hotbar slot to put it there.
- Shift-click an item to send a full stack straight to your hotbar.
- Drop items on the **destroy slot** to delete them; Shift-click it to empty your whole inventory.

Flying and building

- Double-tap Space to take off. Hold Space to rise, Shift to sink, and Ctrl to fly faster. Double-tap Space again, or land, to stop flying.
- Hold the left mouse button to keep breaking blocks quickly. Swords can't break blocks in Creative, so you can fight safely near your builds.
- Middle-click a block in the world to put that block in your hand.
- You can reach 5 blocks in Creative (4½ in Survival).
- **Spawn eggs** (Spawn Items tab): right-click the ground to place a pig, cow, sheep, chicken, goat, rabbit, Fellhound, villager, Sentinel, Shambler, Bone Archer, Crawler or Dustwalker. Creatures placed this way are saved with the world.
- The Lumen Block is a glowing block — handy for lighting large builds. In Survival you can only get it from an Expert mason.
- You can switch between Survival and Creative at any time from **Options…** in the pause menu.

11 The world, day and night

Landscapes

The world keeps generating as you travel, in every direction. You'll find:

- **Plains** with tall grass and flowers, and **forests** full of oak trees.
- **Birch forests** of pale birch trees, and **taiga**: cool forests of tall, dark spruce.
- **Deserts** of sand over sandstone, with cacti and dry shrubs.
- **Badlands**: red sand and hills striped with bands of terracotta.
- **Hills, mountains** and **snowy peaks**. Climbing above height 105 earns **Summit**.
- **Oceans, beaches** and **rivers**. Press Space to swim up; watch your air bubbles.
- **Caves**: winding tunnels and large caverns, the best places to find coal and iron. Bring torches.
- **Villages** (if Structures is on) in plains, deserts and taiga (section 16).
- **Ruins** (if Structures is on): broken cobblestone walls with a loot chest.
- **Lava** deep underground, and hidden **dungeons** (section 21).

The world is 128 blocks tall. The sea is at height 62; bedrock lines the very bottom. Press F3 to see your exact position.

Day and night

A full day lasts **20 minutes**: about 10 minutes of daylight, then sunset, about 10 minutes of night and sunrise. The sun and moon cross the sky and stars come out at night.

- Light from the sky fades at night; torches and furnaces keep glowing.
- Hostile creatures appear anywhere dark; they burn when the sun comes up.
- Sleep in a bed to skip straight to morning (section 12), or wait out the night in a lit shelter.
- Rain, snow and thunderstorms come and go (section 17).

Chests

Craft a chest (8 planks) and right-click it to store up to 27 stacks. Chests are saved with the world. Breaking a chest spills its contents onto the ground.

> **Good to know**
>
> Birch forests, taiga, badlands and cacti appear in worlds created with Blockfell 1.1. Worlds made with 1.0 keep their original landscape, but rune ore is now found deep underground in them too. Lava and dungeons appear in worlds created with 1.7. There is no multiplayer in this version.

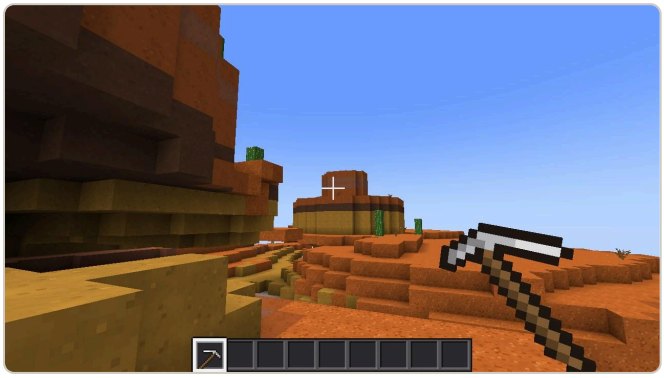
New landscapes in 1.1



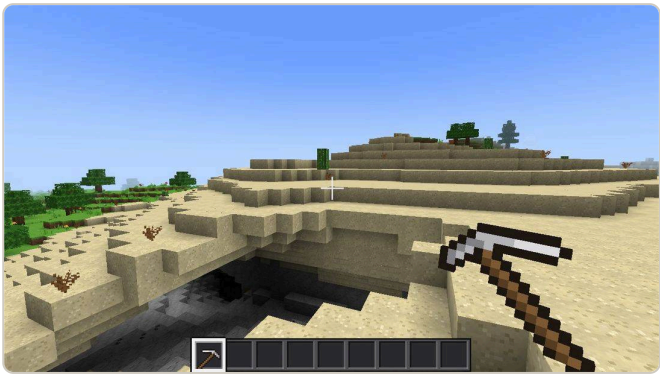
Birch forest. Pale birch trunks; the logs make ordinary planks.



Taiga. Tall spruce trees; home to rabbits, goats and sheep.



Badlands. Layered terracotta hills over red sand. Watch for Dustwalkers at night.



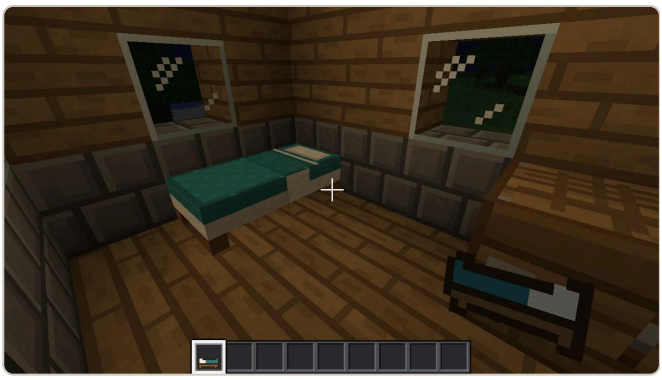
Desert. Now dotted with cacti (don't touch!) and dry shrubs, which drop sticks.

12 Beds, doors, slabs and stairs

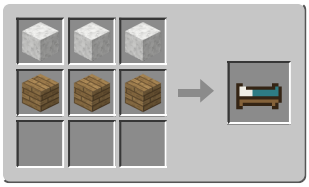
The pieces that turn a shelter into a home. All of them are made on the crafting table from things you already collect.



A cottage with a brick-stair roof, an oak door between two wall torches and a stone-slab path. A stream runs past.

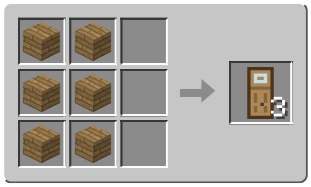


Inside: a bed, a crafting table, a Rune Table and wall torches.



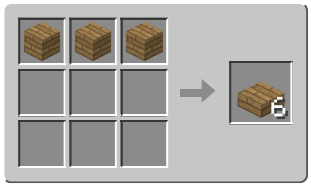
Bed

CRAFTING TABLE



Oak Door ×3

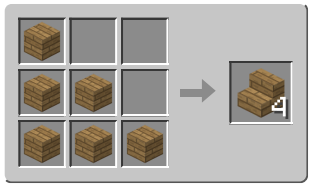
CRAFTING TABLE



Oak Slab ×6

CRAFTING TABLE

Any slab material in a row.



Oak Stairs ×4

CRAFTING TABLE

Any stairs material.

Beds and sleeping

- **Placing:** point at the floor and right-click. The bed stretches away from you, so it needs two blocks of floor.
- **Respawn point:** right-click a bed at any time. “Respawn point set” means that when you die, you come back next to this bed.
- **Sleeping:** once night has fallen, right-click the bed. The screen fades, and about 5 seconds later it is morning. **Leave Bed** gets you up early.
- You can’t sleep while a hostile creature is within about 8 blocks (“You may not rest now”), or from more than 4 blocks away.
- If your bed is broken or blocked in, you respawn at the world’s spawn point instead.

Doors

- A door is two blocks tall and needs a solid block to stand on. It sits on the side of the block nearest to you, and its hinge goes on the side you click.
- **Right-click** either half to open or close it. Hostile creatures can’t open doors (villagers can).
- Place two doors side by side to make a double door.
- Breaking either half removes the whole door and drops one door item.

Slabs

Half-height blocks, 6 from 3 blocks in a row. Made from oak planks, cobblestone, stone, stone bricks, sandstone or bricks.

- Placed on top of a block, or on the **lower half** of a block's side, a slab sits in the bottom half of the space.
- Placed on the underside of a block, or the **upper half** of a side, it sits in the top half — handy for ceilings.
- Put a matching slab onto a slab and the two join into one full block.

Stairs

4 from 6 blocks in a staircase shape. Made from oak planks, cobblestone, stone bricks, sandstone or bricks.

- Stairs face the way you are looking: you walk up them in that direction.
- Placed on the upper half of a side or on an underside, they go in upside down — great for roofs and overhangs.

> **No jumping needed:** you step up onto slabs and stairs (half a block) just by walking. Creatures do too.

Wall torches

Point at the **side** of a block and place a torch: it hangs on the wall, leaning out. On top of a block it stands upright.

- If the block a torch hangs on is removed, the torch drops as an item.
- Flowing water washes torches away.



The same cottage at dusk: the wall torches by the door light the path.

MATERIAL	SLAB	STAIRS
Oak Planks	Oak	Oak
Cobblestone	Cobblestone	Cobblestone
Stone	Stone	—
Stone Bricks	Stone Brick	Stone Brick
Sandstone	Sandstone	Sandstone
Bricks	Brick	Brick

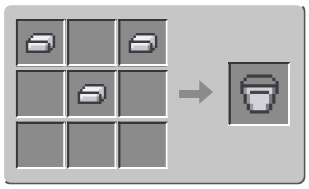
13 Water and buckets

Water now flows: it spreads, runs downhill, pours over edges and dries up when its source is taken away.



A spring on a rock: the water runs to the edge, falls, and flows away along a trench, getting shallower as it goes.

Buckets



Bucket

CRAFTING TABLE

- Right-click a **still water** block (a source, like the sea or a lake) with an empty bucket to fill it.
- Right-click with a **Water Bucket** to pour: it places a new source and gives you the empty bucket back.
- An empty bucket also scoops up a **lava** source: a **Lava Bucket** (section 21).
- In Creative the bucket never empties.

How water flows

- A source spreads up to **7 blocks** across flat ground, getting shallower with each block.
- Over an edge it falls straight down, then spreads again where it lands. On a slope it heads for the nearest way down.
- Take the source away (fill a bucket, or put a block on it) and the stream shrinks and dries up.
- **Endless water:** two sources with one gap between them fill the gap with a third. Dig a 2×2 hole, pour a bucket into two opposite corners, and you can scoop from it for ever.

Water and you

- Currents push you, creatures and dropped items downstream — a stream makes a handy item carrier.
- Hold Space to swim up, even up a waterfall. Watch your air bubbles when under water.
- Landing in water cancels fall damage.
- Flowing water washes away torches, flowers, tall grass and dry shrubs; they drop as items.
- Water flows only near you (in loaded parts of the world). A stream you leave carries on when you come back.

14 Runes

Runes are Blockfell's way of improving your gear. Carve them into tools, weapons, bows and armour at a **Rune Table**, paying with Rune Shards and experience levels.



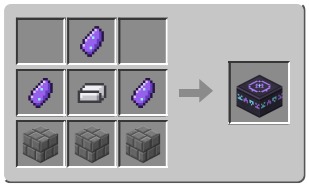
The Rune Table with an iron sword and 12 rune shards. Green numbers are the experience levels needed.



After inscribing: the sword shimmers, and its tooltip lists its runes.

How to inscribe

1. **Find rune ore** deep underground (below about height 30): stone with violet crystals that glow faintly in the dark. Mine it with a **stone pickaxe or better** for 1–3 **Rune Shards**.
2. **Craft a Rune Table** and place it. It gives off a soft light.
3. Right-click the table. Put a tool, sword, bow or piece of armour in the **left** slot and rune shards in the **right** slot.
4. Three offers appear. The number on the right is the **experience level** you need (3, 10 or 20). The small shard shows the **cost**: offer 1 uses 1 shard and 1 level, offer 2 uses 2 of each, offer 3 uses 3. "+1" means a second rune comes with it.
5. Click an offer. The item shimmers and is ready to use. Each item can be inscribed once, and single items only (not stacks).



Rune Table

CRAFTING TABLE

Earning levels

Experience comes from mining ore (coal, iron and rune ore), taking smelted items out of a furnace, and defeating creatures. Save up to level 20 for the best offers.

Offers for an item stay the same until you inscribe something; then new ones are dealt.

The eight runes

RUNE	GOES ON	RANK I	RANK II	RANK III
Swift	Pickaxe, axe, shovel	Mines faster (+2)	+5	+10
Sturdy	Everything	Wears out 2× slower	3× slower	4× slower (armour gains less)
Keen Edge	Sword, axe	+1 attack damage	+1½	+2
Plunder	Sword	Creatures may drop 1 extra	up to 2 extra	up to 3 extra
Bounty	Pickaxe	Ores may drop up to 2×	up to 3×	up to 4×
Might	Bow	Arrows +50% damage	+75%	+100%
Warding	Any armour	−4% damage taken	−8%	−12% (pieces add up, to at most −80%)
Soft Landing	Boots	−12% fall damage	−24%	−36%

Which runes go where

ITEM	POSSIBLE RUNES
 **Pickaxe**	Swift, Sturdy, Bounty
 **Axe**	Swift, Sturdy, Keen Edge
 **Shovel**	Swift, Sturdy
 **Sword**	Keen Edge, Sturdy, Plunder
 **Bow**	Might, Sturdy
 **Helmet, chestplate, leggings**	Warding, Sturdy
Boots	Warding, Sturdy, Soft Landing

Tips

- Inscribe your best gear: an iron pickaxe or sword is worth more runes than a wooden one, and each item can only be inscribed once.
- A **Bounty** pickaxe pays for itself: it works on coal, iron and rune ore, so you get more shards for the next inscription.
- **Warding** on every piece of armour adds up. Add **Soft Landing** boots before exploring mountains.
- Not happy with the offers? Inscribe something cheap first (a wooden shovel with offer 1): that deals new offers for everything.
- Runes are kept when you save, and travel with your world in backups.

15 Farming

Grow your own food. Till the soil with a hoe, plant seeds or carrots, keep the field watered and lit, and harvest when the crops are ripe.



A 9×9 field: wheat on the left, carrots on the right, a water channel down the middle. Wheat turns golden when ripe. Hay bales and a Grain Bin on the left.

Getting started

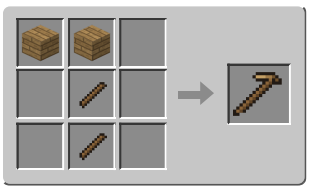
1. **Get seeds:** break tall grass — about one in eight drops **Wheat Seeds**. Village chests and farmers are good sources too.
2. **Craft a hoe** and **right-click** the top of a grass or dirt block to turn it into **farmland**.
3. **Plant:** hold seeds or a **Carrot** and right-click the farmland.
4. **Wait:** crops grow on their own while the chunk is near you. Wheat has 8 stages, carrots 4.
5. **Harvest:** break the ripe crop. Ripe wheat gives 1 wheat and 1–3 seeds; ripe carrots give 2–4 carrots. Unripe crops only give back what you planted.

Growing well

- **Water:** farmland within 4 blocks of water (same level or one above) becomes darker, **moist** farmland. Crops on it grow twice as fast.
- **Light:** crops need light level 9 or more — daylight, or torches for an indoor farm.
- **Bone meal** makes a crop jump ahead a few stages at once. On grass, it sprouts tall grass and flowers around it.
- **Don't jump on farmland:** landing on it tramples it back into dirt. Walk across it, or lay a path.
- Dry farmland with nothing planted slowly turns back into dirt.
- A **shovel** right-clicked on grass makes a **village path**.

What to do with it

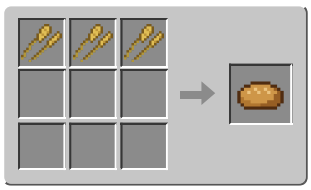
Bread	Restores 2½ drumsticks.
Carrot	Eat raw (1½ drumsticks), or plant more.
Hay Bale	Stores 9 wheat in one block. Landing on hay takes 80% less fall damage.
Amber	Farmers pay amber for wheat, carrots and seeds (section 16).



Hoe

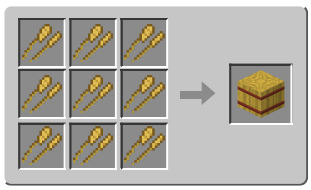
CRAFTING TABLE

Also in stone and iron.



Bread

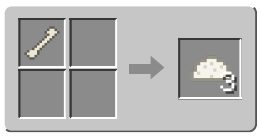
CRAFTING TABLE



Hay Bale

CRAFTING TABLE

Put it back in the grid for 9 wheat.



Bone Meal ×3

INVENTORY OR TABLE

16 Villages and villagers

Villages are small settlements of houses around a well, joined by paths, with a farm and lamp posts. Villagers live, work and trade there — and need your help when raiders come at night.



A village in the plains: timber houses with oak and brick roofs around a well.



Villagers by the well. Their clothes show their job: a farmer's straw hat, a smith's apron, a scribe's violet robe.

Finding a village

- Villages are generated in worlds created with Blockfell 1.2 with **Structures** on. They are fairly rare — at most one in each area of about 300 × 300 blocks — and sit on dry, fairly flat **plains**, **desert** or **taiga**.
- They take on the local style: **timber** in the plains, **sandstone** in deserts, **spruce and stone** in the taiga.
- The bigger houses keep a chest with some bread, seeds, torches, amber and other useful things.

A villager's day

- **Morning and afternoon:** villagers wander the streets and spend time at their **workstation**. Farmers harvest ripe crops and replant.
- **Midday:** they meet round the village bell or the well (section 26). In the rain they stay indoors.
- **Evening:** everyone walks home, opening and closing the door behind them.
- **Night:** they sleep in their own bed until sunrise.
- They run from hostile creatures and hurry indoors during a raid.
- Each dawn, a village with at least two villagers and a free bed welcomes a newcomer.

Professions

A villager's job comes from the workstation it has claimed. A villager without a job takes any free workstation within about 16 blocks — so you can give jobs yourself by placing one.

WORKSTATION	JOB	TRADES IN
Grain Bin	**Farmer**	Wheat, carrots and seeds for amber; bread, apples, bone meal, cooked meat
Forge	**Smith**	Coal and iron for amber; iron tools, armour, buckets; rune-inscribed gear
Mason's Bench	**Mason**	Stone and cobblestone; bricks, glass, terracotta; Lumen blocks
Scribe's Desk	**Scribe**	Feathers, leather, string, bones; rune shards, a Rune Table, rune-inscribed armour
Fletching Bench	**Fletcher**	Sticks, flint and feathers; arrows and bows
Map Table	**Mapmaker**	Leather, feathers, string; maps to ruins, dungeons and villages (section 26)

Break a workstation and a villager who has never traded goes back to having no job. Once it has traded, it keeps its profession.

Trading



A farmer's trades. Left: what you give; right: what you get. The bar under the title fills as the villager gains experience.

1. **Right-click** a villager that has a job. (One without a job just tells you so; a village that distrusts you won't trade.)
2. Each row is an offer: the items on the left buy the item on the right. Hover over a row to see the details.
3. **Click** an offer to trade once; **Shift**-click to trade as many times as you can afford.
4. Press **Esc** or **E** to close.

> **Amber**
>
> **Amber** is the villagers' money. Sell them things they need (wheat, coal, stone, feathers…) for amber, then spend it on things you need.

Levels and restocking

- Every villager starts as a **Novice** with two offers. Trading gives it experience; it becomes **Apprentice**, **Journeyman** and finally **Expert**, learning two new offers each time.
- Each offer can only be used a set number of times, then shows **SOLD OUT**. The villager **restocks** when it next works at its workstation (at most every few minutes).
- Expert smiths and scribes sell iron gear that already carries runes.
- Levels, offers and professions are saved with the world.

Growing a village

Build a house with a **bed** in or next to a village: at dawn a newcomer moves in (one a day, while the village has at least two villagers). Place a **workstation** nearby to give it a job. In Creative, a villager spawn egg puts a villager anywhere; it finds a free bed and workstation within about 16–20 blocks.

The Sentinel



A Sentinel and a villager at the edge of the village.

- Every village has a **Sentinel**, a tall stone guardian with 50 hearts. It patrols the streets and attacks any hostile creature within 16 blocks, flinging it into the air.
- It leaves you alone — unless you hit it or a villager. Then it stays angry for about half a minute and hits hard. It barely moves when struck.
- If a village with at least three villagers has lost its Sentinel, a new one arrives at dawn.
- A Sentinel drops 3–5 iron ingots.

Night raids



A raid at dusk: the bar at the top shows the wave and how much of it is left.

If you are in a village when night falls, raiders may come: a horn sounds and a **raid bar** appears. The chance is 15% on Easy, 25% on Normal and 40% on Hard, never on Peaceful, and at most every other night for each village.

- Raiders come in **waves** (2 on Easy, 3 on Normal, 4 on Hard) from outside the village: Shamblers (Dustwalkers around desert villages), Bone Archers and Crawlers.
- They march on the village and go after villagers and the Sentinel as well as you. Villagers run home and stay awake until it is over.
- Defeat every wave to win. If every villager dies, the raid is lost. Walking far away for about half a minute also ends it.

Victory

The villagers give you a gift of amber and bread, you earn **Hero of the Village**, and your standing there rises by 40, so its traders charge you less (section 26).

Defending a village

- Light the streets and fix broken doors before dark.
- Fight near the Sentinel — it does a lot of damage.
- Deal with Bone Archers first; they pick off villagers from a distance.
- Armour and a rune-inscribed sword make raids much easier.

Villages in other lands



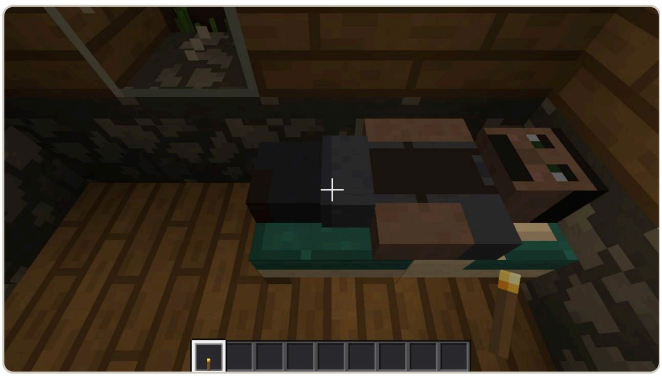
Desert village. Sandstone houses with flat roofs.



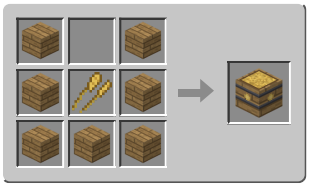
Taiga village. Spruce frames with cobblestone and stone-brick roofs.



A village farm. The farmer harvests ripe wheat and carrots and plants new ones.



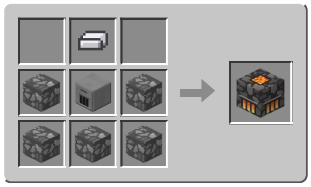
At night every villager sleeps in its own bed.



Grain Bin

CRAFTING TABLE

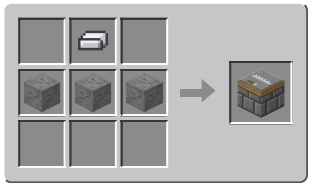
Farmer



Forge

CRAFTING TABLE

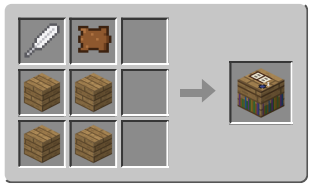
Smith



Mason's Bench

CRAFTING TABLE

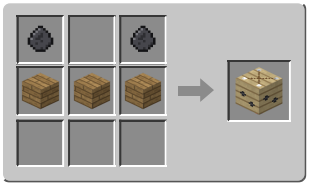
Mason



Scribe's Desk

CRAFTING TABLE

Scribe



Fletching Bench

CRAFTING TABLE

Fletcher

> **Good to know**
>
> Villages only appear in worlds created with Blockfell 1.2 (with Structures on). Worlds from 1.0 and 1.1 keep their landscape unchanged, so they have no villages — but farming works in every world, and in Creative a villager spawn egg, a bed and a workstation are all it takes to start a village of your own.

17 Weather

Skies don’t stay blue for ever. Clear spells of about 8 to 30 minutes are broken by showers of 3 to 8 minutes, and now and then by a thunderstorm.



Rain in a village. The sky turns grey, the light dims and raindrops splash on the ground. Roofs keep it off.



A thunderstorm: darker still, with lightning striking the ground.

Rain, snow and dry lands

- **Rain** falls in most places. It stops at the first solid block above you, so a roof or an overhang keeps you dry.
- **Snow** falls instead in snowy lands and high up in the mountains.
- **Deserts and badlands** stay dry: the sky clouds over, but nothing falls.
- You’ll hear the rain, muffled when you’re indoors.

Thunderstorms

- About one spell of bad weather in three is a thunderstorm.
- **Lightning** strikes the highest block of a random spot near you every ten seconds or so. Anything within a few blocks of a strike takes about 2½ hearts of damage. Stay indoors, or keep moving.
- Seeing a strike close by earns **Storm Watcher**.



Snow drifting down over snowy lands.

What the weather changes

Crops	Rain waters farmland and crops under the open sky: they grow as fast as moist farmland next to water.
Shamblers and Bone Archers	Don’t burn in the rain: a wet morning won’t clear them away.
Thunderstorms	Dark enough for hostile creatures to appear in the open, even at midday.
Sleeping	You can sleep during a thunderstorm, even by day. You wake the next morning to clear skies.

Controlling the weather

- **Weather Cycle** (Options in the pause menu, or Game Rules when creating a world) — turn it off to keep today’s weather for good.
- In Creative, the **Weather** button in Options switches straight to clear, rain or thunderstorm.
- The weather is saved with your world.

18 Fellhounds

Fellhounds are wild, wolf-like hounds of the northern woods: grey-brown with a dark back, a pale ruff and amber eyes. Win one over and it will follow you anywhere.



A wild pack at the edge of the woods.



Two tamed Fellhounds, with their blue collars: one sitting and staying, one ready to follow.

Wild Fellhounds

- They roam forests, the taiga and snowy lands in packs of two to four, and chase rabbits.
- They leave you alone — but **hit one and the whole pack turns on you** for about half a minute.

Taming one

1. Hold raw meat: **Raw Beef**, **Raw Porkchop**, raw mutton, raw chicken or raw rabbit.
2. Right-click a wild Fellhound that isn't angry. Each piece has a one-in-three chance; a puff of smoke means "not yet".
3. Hearts and a blue collar: it's yours. You earn **Loyal Companion**.

> **Tips**
>
> Tame two or three: a small pack makes short work of a Shambler. Make them sit indoors before you go mining, or take them along — but keep meat handy to heal them after a fight.

Your companion

Follows you	It keeps close and finds its way round obstacles. It can't open doors.
Catches up	Left over 14 blocks behind, it bounds to your side (not while you fly).
Sit / follow	Right-click it to make it **sit and stay**; right-click again to call it.
Fights for you	It attacks whatever you attack or attacks you (never villagers or the Sentinel). Its kills give you experience.
Health	10 hearts. Its **tail droops** as it gets hurt. Feed it any meat, raw or cooked, to heal it.
Safe from you	Your own swings and arrows never hurt it.
Saved	Tamed hounds never wander off, and are saved with the world.



A Fellhound takes on a Shambler that came for its owner.

19 Decorating your home

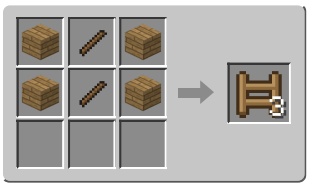
Fences and gates, ladders, trapdoors, glass panes, sixteen colours, signs, lanterns, flower pots and paintings: everything that turns four walls and a roof into a home. All of it is crafted from things you already collect, and it all sits in the **Decoration Blocks** tab in Creative.



Rose Cottage: a fenced garden with a gate, lantern posts, flower beds, potted flowers, glass pane windows and a sign.

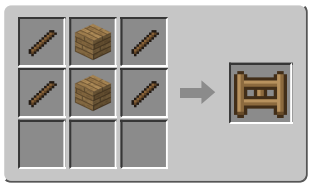


Inside: a wool rug, three paintings, a lantern on a fence post, a hanging lantern and a ladder up to a trapdoor in the roof.



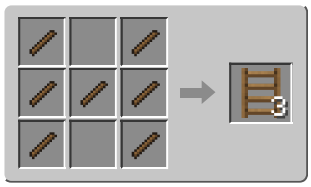
Oak Fence ×3

CRAFTING TABLE



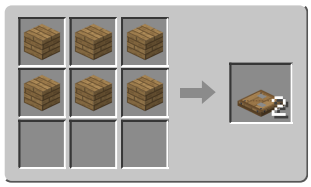
Oak Fence Gate

CRAFTING TABLE



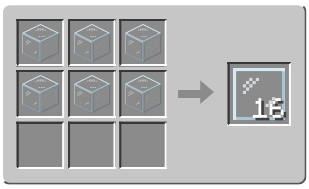
Ladder ×3

CRAFTING TABLE



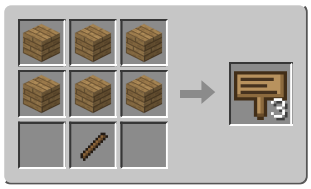
Oak Trapdoor ×2

CRAFTING TABLE



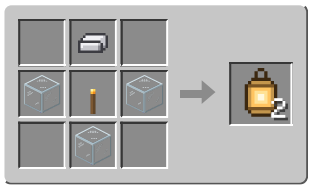
Glass Pane ×16

CRAFTING TABLE



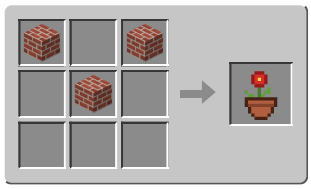
Oak Sign ×3

CRAFTING TABLE



Lantern ×2

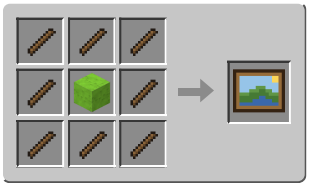
CRAFTING TABLE



Flower Pot

CRAFTING TABLE

Or any terracotta.



Painting

CRAFTING TABLE

Any colour of wool.

Fences and gates

- Fences **join up** with the fences, gates and solid blocks next to them.
- They are **1½ blocks tall**: nobody can jump over them — not you, not animals — so they make perfect pens and gardens. Creatures never appear on top of them.
- A **gate** fits in a gap in the fence. Right-click to open or close it: it always swings away from you. Walk through while it's open.
- Hostile creatures and villagers can't open gates (villagers do open doors).

Ladders

- Click the **side** of a solid block to put a ladder on it. Stack them up a wall.
- **Walk into** a ladder (or hold Space) to climb. Let go to slide gently down — no fall damage.
- Hold Shift to stop and stay where you are.
- A ladder drops off if its wall is removed.

Trapdoors

- A hatch one block wide. Placed on a floor, or on the lower half of a block's side, it lies flat at the bottom of the space; on the upper half or an underside, at the top.
- Right-click to open: it swings up against its hinge. Stand on a closed one; open it and drop through. Great over a ladder shaft.

Glass panes

- Thin glass, 16 from 6 glass blocks. Panes join each other, plain glass and solid blocks into window frames.
- They let all the light through and stop you like a wall. Like glass, they break without dropping anything.

Lanterns

- The brightest light you can make (as bright as a Lumen Block). Stand one on any solid block — a fence post makes a fine lamp post.
- Click the **underside** of a block and it hangs from a short chain.
- New villages (worlds created with 1.4) light their streets with lantern posts.

Flower pots

- Put a pot down, then right-click it holding a flower, a cactus or a dry shrub to plant it.
- Right-click with an **empty hand** to take the plant back out. Breaking a potted plant gives you both.
- Masons sometimes sell flower pots.



Rose Cottage at night: lanterns light the gate, the porch and the garden.

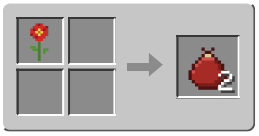
Dyes and coloured wool

Sixteen colours, from flowers and a few other things. Every dye can be crafted in your inventory grid.

TAKE	GIVES
Ember Poppy	**Red Dye** ×2
Sunbell	**Yellow Dye** ×2
Skybell	**Blue Dye** ×2
Moon Daisy or **Bone Meal**	**White Dye** (×2 from the daisy)
Coal	**Black Dye**
Dry Shrub	**Brown Dye**
Cactus in a furnace	**Green Dye**

All four flowers grow on grassland. Skybells and Moon Daisies grow wild only in worlds created with 1.4 — but **bone meal on grass** sprouts all four kinds in any world.

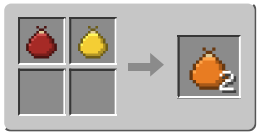
MIX TWO DYES	FOR TWO OF
Red + Yellow	**Orange Dye**
Red + White	**Pink Dye**
Red + Blue	**Purple Dye**
Purple + Pink	**Magenta Dye**
Blue + White	**Light Blue Dye**
Blue + Green	**Cyan Dye**
Green + White	**Lime Dye**
Red + Green	**Brown Dye**
Black + White	**Gray Dye**
Gray + White (or Black + 2 White for 3)	**Light Gray Dye**



Red Dye ×2

INVENTORY OR TABLE

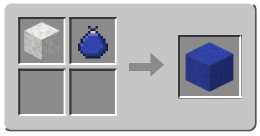
Any flower: its own colour.



Orange Dye ×2

INVENTORY OR TABLE

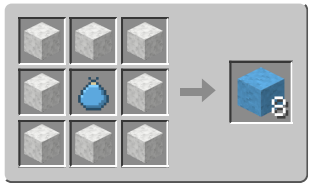
Two dyes mix (table below).



Blue Wool

INVENTORY OR TABLE

Any wool, any dye.



Light Blue Wool ×8

CRAFTING TABLE

Eight at once.



Eight white wool around a light blue dye make eight light blue wool.

Wool

- Put any wool and a dye together for wool of that colour — you can re-dye coloured wool too.
- Eight **white** wool around one dye make eight at once.
- Coloured wool builds like any block: rugs, flags, awnings, pixel art.
- A dye on a sign colours its writing (below).

> Crafting your first dye or dyed wool earns **A Splash of Colour**.

Signs

- Click the **top** of a block: the sign stands on a post, turned to face you. Click the **side** of a block: it hangs flat on the wall.
- The editor opens straight away. Type up to **four lines**; Enter, Tab or Down moves to the next line. **Done** or Esc finishes.
- Right-click a sign to change what it says. Right-click it holding a **dye** to colour the writing. To put a block against a sign without opening it, hold Shift while you place.
- Your first message earns **Signpost**. Signs and their words are saved with the world.



The sign editor...



...and the sign at the garden gate.

Paintings



All eighteen paintings, each one painted just for Blockfell.

- Right-click the **side of a wall**. It picks, at random, one of the **largest** pictures that fit around the spot you clicked: every square needs a solid block behind it and free space in front, and paintings never overlap.
- Hold Shift while hanging it for a **small** one instead.
- **Right-click** a painting to change it to another picture of the same size.
- **Hit** it to take it down. It also falls if its wall is broken or a block is put in front of it.
- Scribes sometimes sell paintings. Your first one earns **Art Lover**.

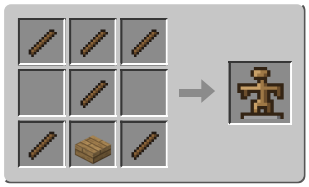
SIZE	PAINTINGS
1 × 1	Orchard Apple, Night Watch, Blue Vase, Toadstool
2 × 1 · 1 × 2	Homeward Sail, Rolling Hills, Hearth Hound · The Lighthouse, Lone Pine, The Falls
2 × 2	Mountain Lake, Village at Dusk, The Scribe, Wildflower Meadow
4 × 2 · 4 × 3	Harvest Moon, Storm at Sea · Northern Lights, River Valley

Armour stands



An armoury: iron, dyed leather and plain leather on four stands, with a sword, a bow, a fishing rod and a torch in their hands.

- Right-click the floor with an **Armour Stand**: it stands there, turned to face you.
- Right-click it holding **armour** and the piece goes on in its place — a cap on its head, boots on its feet — whatever part you aim at. Anything else (a sword, a bow, a torch) goes in its **hand**.
- With an **empty hand**, right-click the piece you want back: aim at its head for the helmet, its feet for the boots.
- **Sneak** (Shift) and right-click with an empty hand to **swap your whole set** with the stand's — a quick change before a trip below.
- **Hit it twice** to knock it down: it drops itself and everything on it. It falls if the floor goes and burns in lava.
- Your first dressed stand earns **On Display**.



Armour Stand

CRAFTING TABLE

Or a stone slab.

> **On a phone or tablet**
>
> Tap the stand to put a piece on or take one off. Touch and hold to knock it down.

Dyed leather armour

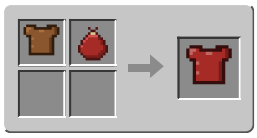


A Leather Tunic with red and yellow dye: an orange tunic.



Your inventory shows the armour you wear on the picture of you.

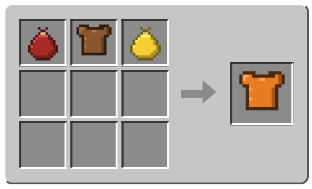
- Put a leather piece and a **dye** anywhere in a crafting grid: out comes the piece in that colour. Sixteen dyes, sixteen colours.
- Two or more dyes (up to eight) **mix** into an in-between colour.
- A dyed piece keeps its wear and its runes. Dye it again any time.
- A dyed piece on its own in the grid comes out **plain leather** again.
- Iron armour can't be dyed. Your first dyed piece earns **Dressed to Impress**.



Red Tunic

INVENTORY OR TABLE

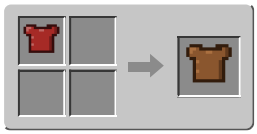
Any leather piece and a dye.



Orange Tunic

CRAFTING TABLE

Two or more dyes mix.



Washed plain

INVENTORY OR TABLE

A dyed piece on its own.

20 Boats and fishing

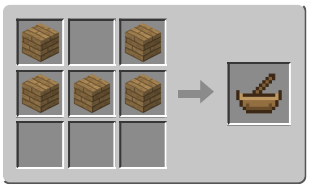
Row across lakes, down rivers and along the coast in a boat you build yourself, and sit by the water with a fishing rod for trout and perch — and now and then something more surprising on the end of the line.



Two boats on the water, ready to climb into.

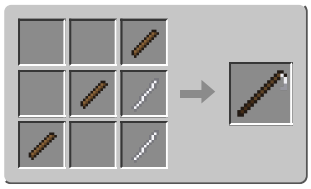


Rowing along the shore. The oars dip as you go.



Boat

CRAFTING TABLE



Fishing Rod

CRAFTING TABLE

Boats

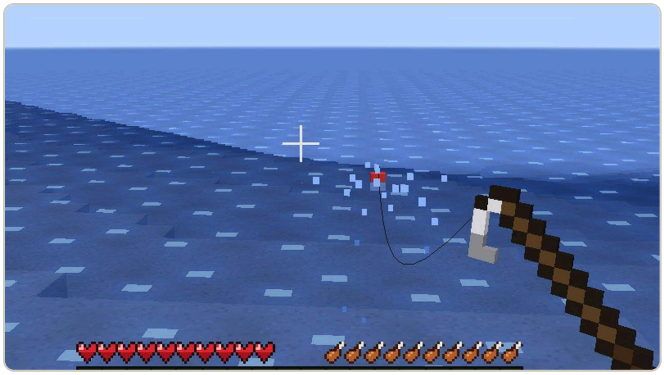
1. Hold the boat and **right-click the water** (tap it on a touch screen). It floats where you point.
2. **Right-click the boat** (tap it) to climb in. You sit in the middle, eyes about a metre above the water.
3. Row away. You can still look around freely while you row.

TO	KEYBOARD	TOUCH
Row forwards / backwards	W / S	Joystick up / down
Turn left / right	A / D	Joystick left / right
Climb out	Shift	Sneak button

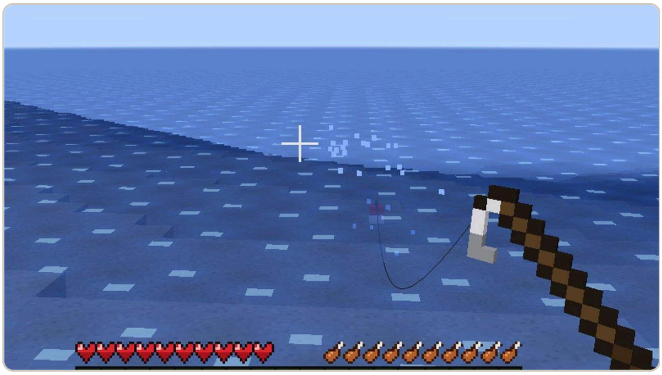
Good to know

- A boat is quicker than sprinting — about 6 blocks a second at full speed — and rowing doesn't make you hungry.
- Currents in rivers push it along gently. The shore stops it; it never climbs onto land by itself.
- On dry land a boat can be placed, but it barely moves when you row.
- Climbing out puts you on dry land beside the boat if there is some, otherwise in the water.
- **Hit a boat three times** in quick succession to break it: it drops as a Boat to pick up and use again. In Creative one hit removes it.
- Save and Quit while in a boat, and you're still sitting in it when you come back.
- Rowing earns **Set Sail**; **Distance by Boat** is counted in Statistics.

Fishing



A fish is coming: follow the trail of ripples to the float.



The bite! The float is pulled under with a splash — reel in now.

Catching a fish

1. Hold the fishing rod, face the water and **right-click** (tap) to cast. The float flies out, lands and bobs on the surface.
2. Wait. After 5 to 30 seconds a fish comes: a **trail of ripples** swims towards the float.
3. When the float is **pulled under** with a splash, right-click (tap) again at once: you have one to two seconds.
4. The catch flies out of the water to you, with a little experience.

- Too slow? The fish gets away, and you wait for the next one. Reeling in early costs nothing.
- Fish **bite sooner in the rain**.
- You can fish from the shore or sitting in a boat.
- The line snaps if you switch to another item or go more than 32 blocks from the float.

What you can catch

CHANCE	CATCH
Fish · 85%	**Raw Trout** (most often), **Raw Perch**
Junk · 10%	A stick or two, string, a bone or leather
Treasure · 5%	**Amber**, a **Rune Shard** or a worn bow

Cooking the catch

Raw fish is barely filling (1 drumstick). Cook it in a furnace: **Cooked Trout** gives 2½ drumsticks and **Cooked Perch** 3, and perch keeps you going especially long.

The rod

- A rod lasts **64 catches**: each catch wears it by one. Reeling in a float that landed on the ground wears it by two. A **Sturdy** rune (section 14) makes it last longer.
- Your first fish earns **Catch of the Day**; every catch counts in **Fish Caught** in Statistics.

21 Lava and dungeons

Deep down the rock runs hot. Below height 11 the caves are flooded with glowing lava, more of it lies buried in pockets in the stone, and here and there a cave runs past a forgotten dungeon, where a Monster Cage keeps making creatures.



A lava lake at the bottom of a cavern. Lava lights up everything around it.



Water poured against a lava pool: wherever it touches the lava, the lava hisses and sets into Cinderstone (the dark blocks).

Lava

- **It glows** as brightly as a Lumen block, and pops and spits embers.
- **It flows slowly**, only 3 blocks from its source (water goes 7), and pours down over edges. Take the source away and it drains.
- **It burns.** In lava you lose 2 hearts every half second and sink slowly, and you are set **on fire** for 15 seconds: half a heart a second. Armour helps a little in the lava itself, but not against the flames.
- **Water or rain puts you out.** Jump into water, or stand in the rain.
- Creatures burn in it too. Items dropped in it are gone for good, and so is a boat. Torches and plants in its way burn up.
- In Creative lava does you no harm.

Where it is

- In **caves below height 11**: the deepest caverns are lakes of lava.
- In **buried pockets** from height 12 to 40, sealed in the rock — until you dig into one. Listen for popping and look for a glow.

No lava? Worlds made before 1.7 have none, so since 1.10 Cinderstone can also be crafted (section 5).

Lava in a bucket

- Right-click a still lava source with an empty bucket: a **Lava Bucket**. Right-click to pour it out again.
- As furnace fuel it smelts **100 items** and hands the bucket back.

When water meets lava

MEETING	MAKES
Water touches a lava **source**	**Cinderstone**
Water touches **flowing** lava	**Cobblestone**
Lava pours down **onto** water	**Stone**

Cinderstone is glassy black rock veined with red, and the hardest block there is: it takes an **iron pickaxe** about 9 seconds, and anything less gets nothing. Pouring water onto lava earns **Cooling Off**, and it is the safest way to cross a lava lake: turn its surface to Cinderstone as you go. Eight of it make a **Deepgate** (section 22).

> **Digging deep**
> Never dig straight down. Carry a water bucket, keep a block in your hand to plug a leak, and step back the moment you see an orange glow through a crack.

Dungeons



A dungeon (lit up with lanterns for the picture): mossy cobblestone, two chests and the Monster Cage in the middle.



The cage close up. The little figure inside shows which creature it makes.

Finding one

- Dungeons are small cobblestone rooms buried in the rock between heights 12 and 56, always beside a cave: look for mossy cobblestone in a cave wall.
- There is about one for every 8 to 10 chunks (a chunk is 16 × 16 blocks), in worlds created with 1.7 with **Structures** on.
- Each has a **Monster Cage** in the middle and one or two chests against the walls. Opening one earns **Dungeon Delver**.

The Monster Cage

- It wakes when you come within **16 blocks** (flames flicker inside) and makes its creature — a Shambler, Bone Archer or Crawler — every 10 to 40 seconds, up to 4 at a time and never more than 6 nearby.
- It only works in the **dark**. Light the room with torches (light above 9 all round) and it stops.
- Nothing appears on **Peaceful**.
- Break it with any pickaxe for a burst of experience (the cage itself doesn't drop) and **Cage Breaker**.
- In Creative, use a spawn egg on a cage to change what it makes.

Dungeon loot

HOW OFTEN	WHAT
Common	**Bone**, **String**, spoiled flesh, bread, **Coal**, **Iron Ingot**, arrows
Less common	**Amber**, **Rune Shard**, wheat, apples, a bucket, a bow
Rare	An iron helmet or chestplate; an iron pickaxe or sword already carved with a rune

Clearing a dungeon

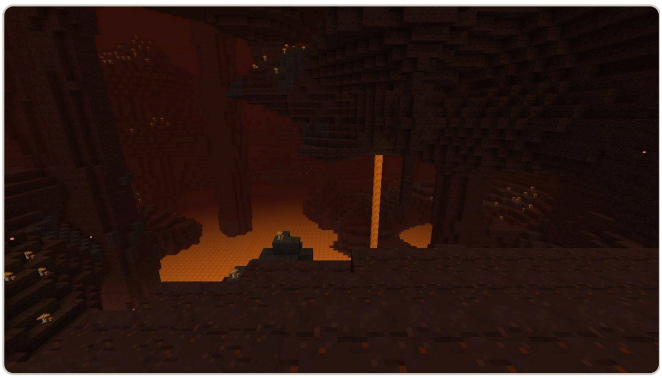
Go in with a sword, armour and food. Deal with the creatures already there, then break the cage straight away — or ring it with torches first so it stays quiet while you empty the chests. A cage you leave switched off makes a safe creature farm you can light and darken as you please.

22 The Cinderdeep

Far below the world lies the Cinderdeep: great caverns of dark Ashrock over a sea of lava, lit by glowing fungus and roamed by creatures made of fire. You get there through a **Deepgate** you build yourself — in any world, old or new.



A lit Deepgate: eight Cinderstone round a glowing middle. Stand in it to go down.



A cavern of the Cinderdeep: the lava sea below, a lavafall, Glowcaps on the ledges and a red haze in the distance.

Building a Deepgate

1. Get **8 Cinderstone** (Cinderstone Bricks work too): from water on lava (section 21) or crafted from stone, coal and iron (section 5).
2. Lay them in a **ring** on the ground: a 3 × 3 square, all at the same height, with the middle left open and something solid under the middle.
3. Hold a **Torch** or a **Lava Bucket** and right-click the floor in the middle. The gate lights up (the torch is used; the bucket comes back empty). A torch already standing in the middle lights it when you finish the ring.

Going through

- Step down into the glowing middle and **stay for 3 seconds** (1 in Creative; flying just above it works too) while the glow builds up. Step out and it fades.
- You come out at the **same place on the map** down below, beside a gate that leads back. If there is no lit gate within 16 blocks, one is made for you on the nearest cavern floor.
- To come home, stand in that gate: you come out beside the one you went in by. Your first trip earns **Into the Cinderdeep**.
- A gate goes out if a block of its ring, or the block under it, is broken. Light it again with a torch.

Down there

- **No sky**: no sun, no night and no weather. A dull red glow lets you see a little; lava, Glowcaps and torches light things properly.
- The **lava sea** fills every cavern below height 25, and lavafalls pour from the walls. Watch your step on ledges.
- **Water won't stay**: poured out, it hisses away as steam. With no water or rain, fire burns until it goes out by itself.
- **Beds don't work** (it is far too hot to sleep) and **maps** only work on the surface.
- **If you die** down here you wake up on the surface, at your bed or spawn point. What you dropped stays where you fell (unless it fell in the lava).
- The Cinderdeep is part of your world: it is saved with it, in backups and in Dropbox. Each world's is different, made from its seed.

> **Before you go**
>
> Take an iron pickaxe, a sword, armour, food, a stack of blocks for bridges and plenty of torches. Note where your gate is down there: it is your way home.

Creatures and treasure



A Smoulderer (left) and a Cinderling on an ash floor, with Glowcaps and an Ember Lamp.



An Ember Shrine: an Ashrock Brick hall with a Monster Cage in the middle and a chest in the corner.

Creatures

CREATURE	HEARTS	WHAT IT DOES
Cinderling	5	Small and quick. Leaps at you from a few blocks away; its touch sets you on fire. Drops Ember.
Smoulderer	12	Tall and hooded. Keeps 5 to 10 blocks away and throws burning embers. Drops Ember, sometimes a Fire Opal.

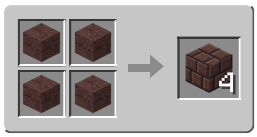
- Both are **fireproof**: lava and flames don't hurt them. Use a sword or a bow.
- They appear wherever it isn't brightly lit — more than a couple of blocks from a torch, lamp or lava — at any time; none on Peaceful. The only animals are Ashboars (next page).

Ember Shrines

- Ruined halls of Ashrock Bricks on the cavern floors, about one in every 9 chunks, with Ember Lamps on their Cinderstone corner posts.
- A **Monster Cage** of Cinderlings or Smoulderers stands in the middle. Fire creatures don't mind light, so torches won't stop it: break it.
- One or two chests hold Ember, iron, Rune Shards, Amber, Glowcaps, sometimes a Fire Opal and, rarely, a Cinder Charm. Opening one earns **Shrine Raider**.

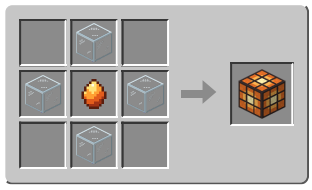
New blocks and items

Ember Ore	Glows in the rock. Any pickaxe: 1–3 **Ember**, the best fuel (24 items).
Fire Opal Ore	Rare, low down near the lava sea. Iron pickaxe: 1 **Fire Opal**.
Glowcap	Glowing fungus. Pick it and plant it on any solid block.
Glimmerstone	Glows on cavern roofs. Breaks into **Glimmer Dust**; 4 make it again.
Ashrock **Ash**	The rock (quick to dig) and dust of the Cinderdeep.



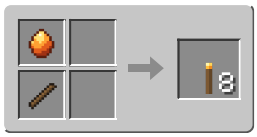
Ashrock Bricks ×4

INVENTORY OR TABLE



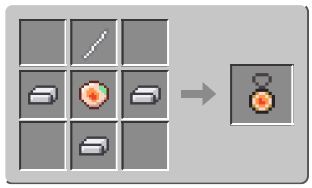
Ember Lamp

CRAFTING TABLE



Torches ×8

INVENTORY OR TABLE



Cinder Charm

CRAFTING TABLE

The Cinder Charm

Carry it in your **off hand** (F swaps it in): fire and lava hurt you half as much, and flames on you go out sooner. Making one earns **Fireproof**.

Ashboars



A herd of Ashboars with a piglet, among the Glowcaps. Their ember seams glow in the dark.



Two Ashboars fed a Glowcap each, and the piglet they had.

- Heavy beasts that roam the cavern floors in **herds of two to four**. Lava and fire don't hurt them.
- **Calm until you hurt one.** Then every grown-up nearby charges, and a hit **tosses you into the air**. Piglets just run away. On Peaceful they never charge.
- **Ember Lamps scare them.** They won't come within about 6 blocks of one, and a charging herd turns back at that distance: light your base and bridges with lamps.
- They drop **Raw Ashboar** and sometimes **Leather**. Cook the meat in a furnace: **Roast Ashboar** is as filling as steak. Fellhounds like it too.

Piglets

- Give two grown-ups a **Glowcap** each (right-click, or tap): hearts appear, they walk together and a **piglet** is born. Your first earns **Ember Piglet**.
- The parents wait 5 minutes before they can have another.
- A piglet grows up in 10 minutes. Each Glowcap you give it takes a minute off.

ASHBOAR	
Hearts	10 (a piglet 5)
Charge	1 heart on Easy, 2 on Normal, 3 on Hard
Where	Cinderdeep cavern floors, 24–48 blocks from you; never near an Ember Lamp

23 The Starhollow

High above the world, islands of pale Starstone float in a sky full of stars. Their guardian, the **Hollowdrake**, circles over its Roost. You get there through a **Stargate** — in any world, old or new.



A lit Stargate on the surface: eight Glimmerstone round a violet glow. Stand in it to rise up.



Arriving: the four pylons with their Storm Bells, the Roost in the middle and the Hollowdrake above.

Lighting a Stargate

1. Get **8 Glimmerstone**: break the glowing clusters on the Cinderdeep's cavern roofs for **Glimmer Dust**; 4 dust make a block.
2. Make a **Star Lens** (recipe on page 3 of this section) from 2 Rune Shards, 2 Glimmer Dust and a Fire Opal.
3. Lay the Glimmerstone in a **ring** on the surface, like a Deepgate: a 3 × 3 square with the middle open and something solid under it.
4. Right-click the middle with the Star Lens. The lens is used up and the gate lights. (A torch won't do, and the stars are too far away from the Cinderdeep.)

Going up and coming home

- Stand in the glow for **3 seconds**. Every Stargate leads to the same place: the **arrival gate** on the main island, facing the Roost. Your first trip earns **Among the Stars**.
- The arrival gate takes you home, beside the Stargate you came up by.

Up there

- It is **always night**, clear and starry: no sun, no rain, no clouds. You can't sleep and maps don't work.
- The **main island** is about 130 blocks across. In the middle is the **Roost**, with four tall **pylons** round it, each with a ladder up the side facing the Roost and a **Storm Bell** on top.
- Beyond a wide gap of empty sky lie the **outer islands**, with glowing **Starbloom** and now and then a **Starfall shrine**: a chest of Glimmer Dust, Drift Silk, rune shards and amber, sometimes a Star Lens or a Star Scale. Build a bridge, or glide.
- **Drifters** float gently between the islands. They never attack; they drop Drift Silk.

> **Falling off**
>
> There is nothing below the islands but sky. Fall off and the stars **catch you and carry you home** to your bed or spawn point — with everything you had. You don't lose a thing.

The Hollowdrake



The Hollowdrake flies past a pylon, inside its shield. The bar at the top shows its health and how many bells still ring.



A Storm Bell on top of a pylon. Climb the ladder and right-click (or tap) it to silence it.

How to beat it

1. **Silence the four Storm Bells.** While any bell rings, the Hollowdrake is **shielded**: your blows and arrows bounce off. Climb each pylon and right-click its bell. The fourth earns **Silence the Bells** and the shield is gone.
2. **Fight it.** Hit it with a sword when it swoops down, or shoot it with a bow as it flies.
3. **Catch it resting.** Every 35 to 60 seconds it lands on the Roost for about 11 seconds. Hits there do half as much damage again — but watch out for its bite.

> **Before you go**
> Take armour, a sword, a bow with plenty of arrows, food and some blocks. On top of a pylon you are safe from its swoops (it throws star bolts instead).

HOLLOWDRAKE	
Hearts	100
Star bolts	Thrown three at a time: 2 hearts each on Normal
Swoop	3 hearts, and it knocks you up and away
Bite	2½ hearts, only while it rests on the Roost
Difficulty	Easy halves all of this, Hard adds half; on Peaceful it can't hurt you

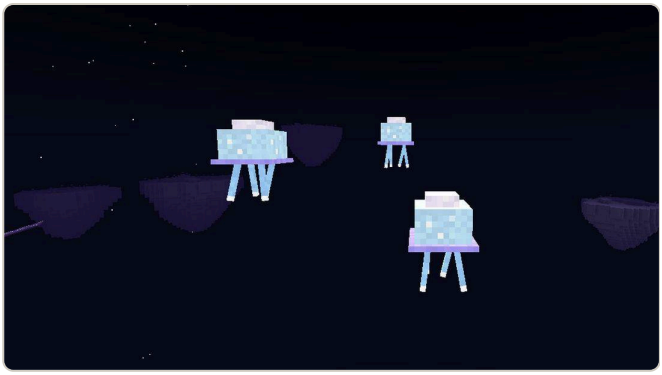
Victory

- Beaten, it rises and fades into starlight, leaving **8 Star Scales**, a heap of experience and a **Hollowdrake Trophy** on the Roost. You earn **Free the Stars**.
- The Roost becomes a **Stargate home**. The first time you step into it you see **the End**: a short poem about your journey and the credits, over the stars. Hold a key or the screen to speed it up, or press Skip. Afterwards, the pause menu has a **Replay the Credits** button: watch it again whenever you like (it plays in place and then takes you straight back to the game).
- Want another fight? Right-click the Roost's gate with a **Star Lens**: the bells ring again and the Hollowdrake returns. **Every win leaves one more Trophy.** Place them as a golden star on a little plinth to keep as a collection (it glows a little, and a pickaxe takes it back).

Starwings, Drifters and the End



Starwings go in the chest slot, and show on the picture of you.



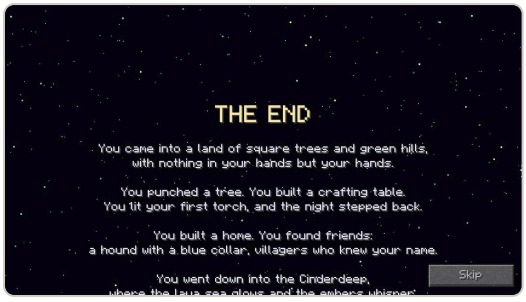
Drifters floating at the edge of the main island.

Gliding

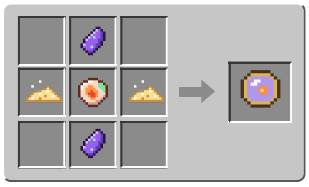
- Make **Starwings** from 6 Star Scales and a Drift Silk, and wear them in the **chest** slot.
- Jump off something high, then **press Jump again in mid-air** (on a tablet, tap the Jump button). You glide! Your first glide earns **Take Wing**.
- Steer by looking. Look down to dive and speed up; look up to slow down and climb a little. Landing, or water, ends the glide — and landing never hurts.
- They wear a little every second (about 7 minutes of gliding). Mend them with Star Scales in the crafting grid.

Drifters

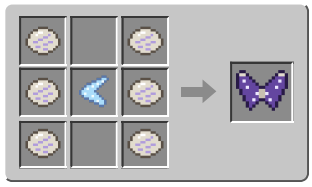
- Gentle, glowing creatures that float about the islands. Hit one and it drifts away; it drops **Drift Silk**.



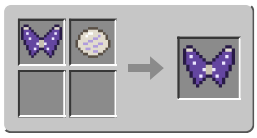
The End: words drift up over the stars.



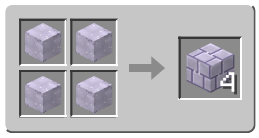
Star Lens
CRAFTING TABLE
Lights a Stargate (used up).



Starwings
CRAFTING TABLE



Mended
INVENTORY OR TABLE
Each scale mends a third.



Starstone Bricks ×4
INVENTORY OR TABLE

Starstone	The pale stone of the islands. Any pickaxe.
Starbloom	Glowing flower. Plant it on solid ground; craft it into 2 purple dye.
Star Scale	Dropped by the Hollowdrake. Makes and mends Starwings.

24 Snow day, pets and farm animals

Have a snowball fight, build a **John** (a snowman helper) to guard your home, and race down a snowy hill on a **Sledgy** (a sledge). Win over a **Tim** (a shy fox cub), give your creatures names, raise young animals and dye your sheep every colour there is.



A John pelts a Shambler with snowballs. Each one knocks it back.



Riding a Sledgy down a snowy hillside, with snow spraying from its runners.

Snowballs

- Break a Snow Block (with anything, even your hand): **4 snowballs**. Four snowballs make a Snow Block again. They stack to 16.
- **Right-click** (tap) to throw one; hold to keep throwing. Your first earns **Snowball Fight!**
- A snowball **knocks a creature back** but doesn't hurt it — except creatures of fire: Cinderlings and Smoulderers take 1½ hearts. It puts out a creature that is on fire.
- Snowballs never hurt you, and they fly past your own pets.

John, the snowman helper

1. Put one Snow Block on top of another.
2. Hold Coal and right-click either block. The snow comes to life! You earn **A Frosty Friend**.

- It throws a snowball about **once a second** at any creature of the night within 10 blocks it can see: a sting of ½ a heart and a big knock back. The creatures it hits come after *it*, not you.
- **Right-click it** to choose: **follow me**, or **guard this spot** (it potters about within 7 blocks).
- It has 4 hearts. It **melts** in water, in deserts and badlands, and in the Cinderdeep — so keep it in cool places. Whatever happens to it, it falls apart into 2–6 snowballs.
- Your blows and arrows pass it by. To take it down, **sneak and hit it**.

Sledgy, the sledge

- Make one (recipe on page 3 of this section), put it down on the ground and **right-click (tap) it** to sit on it.
- **Push off** with W (the joystick forwards). Your feet only help while it is slow — about 3 blocks a second on snow.
- **Steer** with A and D; **brake** with S.
- **Downhill is fast:** every step it drops down on snow turns into speed, up to 12 blocks a second. Going over 6 earns **Downhill Dash**. On the flat it glides a long way, then stops.
- While moving it hops up one-block steps. Off the snow it only drags along slowly; on water it floats.
- **Sneak** to get off. Hit it three times to pick it up again. It burns in lava.

> **Where to find snow**
>
> The **snowy lands** high in the mountains (white grass, spruce trees, falling snow) have the best hills. Snowy grass is as slippery as Snow Blocks — or lay your own run of Snow Blocks down any slope.

Tims (fox cubs) and Name Tags



Wild Tims: a red one from the taiga and a snowy-white one from the snowy lands.



A tamed Tim with its red collar, sitting and staying. A Name Tag put its name above it.

Wild Tims

- Red ones live in the **taiga**, snowy-white ones in the **snowy lands**, alone or in pairs. They pounce about in the snow.
- They are **shy**: walk up to one and it runs away. **Sneak** to creep up on it.

Taming one

1. Hold **Raw Trout** or raw perch (raw chicken or raw rabbit work too).
2. Sneak up and right-click it. Each piece has a one-in-three chance; smoke means “not yet”.
3. Hearts and a red collar: it’s your pet. You earn **Best Friends**.

Your Tim

Follows you	It keeps close, finds its way round things, and bounds to your side if it falls over 14 blocks behind.
Sit / follow	Right-click it to make it **sit and stay**; right-click again to call it.
Presents	Every 5 to 7 minutes or so it brings you a little present: a feather, a flower, an apple, a bone...
Health	8 hearts. Fish, chicken or rabbit heals it (cooked heals twice as much). It is a cub, not a fighter: hurt, it runs to your side.
Safe	Your swings and arrows never hurt it. It never wanders off and is saved with the world.

Name Tags

1. Make a **Name Tag** from string and leather.
2. Hold it and right-click any creature — a pet, a pig, even a Shambler.
3. Type a name (up to 20 letters) and press **Name it** (or Enter). The tag is used up; you earn **What’s Your Name?**

- The name shows above the creature when you are within 14 blocks.
- A named creature **never wanders off** or disappears — even a creature of the night.
- Another tag gives it a new name. Villagers have names already; the Sentinel and the Hollowdrake can’t be named.
- If a named pet dies, the message says its name.
- **Name yourself:** hold a Name Tag and right-click on nothing in particular (no creature, no chest or door). Write your name and your tag is used up. Press **V** (on a tablet, the little **person button** under the pause button) to change the view: **behind you**, **in front of you**, then back to first person. In the third-person views you see your own body with your name floating over your head. The name is saved with your world. Another tag changes it.



Writing a name on a Name Tag. On a tablet, tap the tag to bring up the keyboard.

Farm animals: young ones and coloured sheep



Young ones with the grown-ups: a lamb, a piglet, a calf and a chick. Young animals are small, with big heads.



A flock dyed every colour. The sheep on the right has been shorn.

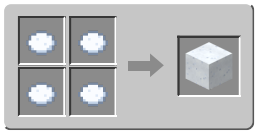
Food and young ones

ANIMAL	ITS FOOD
Pig	**Carrot** or **Apple**
Cow, sheep	**Wheat**
Chicken	**Wheat Seeds**

- **Hold its food** and the animal comes to you, from up to 10 blocks away. Lead it home or into a pen.
- **Right-click** two grown-ups of the same kind with their food: hearts, and they walk together and have a **young one**, with some experience for you. Your first earns **Little Ones**.
- The parents must wait **5 minutes** before they can again.
- A young one follows a grown-up about and grows up in **10 minutes**. Its food makes it grow up a minute sooner. Young ones drop nothing.
- Wild herds sometimes have young ones with them too.
- Feeding an animal never eats the apple or carrot yourself. On a tablet, a tap with its food (or a dye, shears or a Name Tag) uses it on the creature instead of hitting it.

Coloured sheep

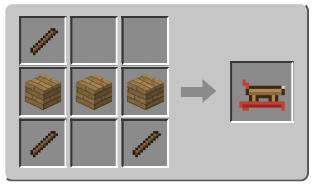
- **Dye:** right-click a sheep with any dye and its wool turns that colour (the dye is used). Your first earns **Woolly Rainbow**.
- **Shears:** right-click a sheep for **1–3 wool** of its colour. It stays bare until it has eaten some grass (the grass turns to dirt); then the wool grows back. A woolly sheep only nibbles.
- **Lambs** take after their parents: two red sheep have a red lamb. Two colours that mix as dyes mix in the lamb: red and yellow make **orange**, red and white **pink**, blue and white **light blue**, red and blue **purple**.
- Wild sheep are mostly white, sometimes light grey, grey, black or brown — and very rarely pink.



Snow Block

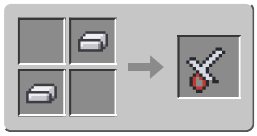
INVENTORY OR TABLE

A Snow Block breaks into 4 snowballs.



Sledgy

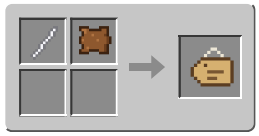
CRAFTING TABLE



Shears

INVENTORY OR TABLE

Wears out after 238 clips.



Name Tag

INVENTORY OR TABLE

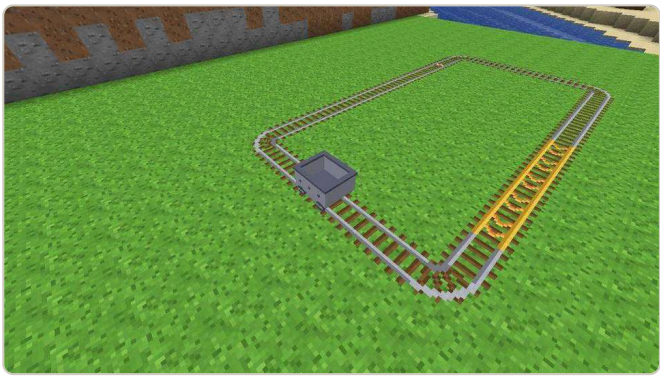
Any two slots.

25 Tracks, carts and parties

Lay iron **rails** over hills and round corners, climb into a **Minecart** and ride. Then throw a party: fill a **Party Bag** and give it to a villager or a pet, pop a **Party Popper**, put up **bunting** and **balloons**, and share out a **Party Cake**.



A loop of track: curves in the corners, gold Boost Rails along the top and a Station Rail (with the red mark) at the bottom.



A Minecart waiting on a curve. Its wheels turn as it rolls.

Laying track

- Rails go on top of any **solid block**. Craft them from iron (recipes below): **Rail**, **Boost Rail** and **Station Rail**.
- You never choose a rail's direction. Each one **looks at the rails round it** and joins them: a straight piece, a **curve** at a corner, or a **slope** when the next rail is one block higher or lower. Lay or break a rail and its neighbours reshape themselves.
- A lone rail runs north–south; a rail with a single neighbour points at it. At a junction the track goes straight on.
- Slopes go up or down **one block at a time**. A curve can't be on a slope: put a flat piece between them.
- Boost and Station Rails are always **straight** — end a curve with an ordinary rail first.
- Rails break quickly with a pickaxe and drop themselves. Flowing water washes them away.

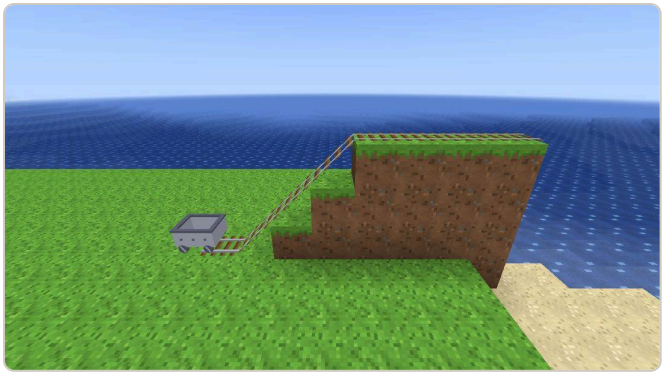
RAIL	WHAT IT DOES
Rail	The ordinary track. Friction slows a cart slowly.
Boost Rail	Gold with chevrons. A cart already rolling is **pushed on** faster and faster, up to about 10 blocks a second. Pushing with W on one gets you going quicker, too.
Station Rail	Red mark. A cart that rolls in is **braked to a stop and held**, even on a hill. Press W to set off again.

Riding a Minecart

1. Hold the **Minecart** and right-click (tap) a rail to put it down. (It only goes on a rail.)
2. Right-click (tap) the cart to **climb in**. Your first ride earns **All Aboard**.
3. Hold W to push off **the way you are looking** along the track. S brakes, and from a standstill backs you up. Shift (Sneak) gets you out.

- **Pushing alone** gets you to about 4 blocks a second. Boost Rails add the rest, up to about **10 a second**. Whizz along a Boost Rail for **Express Service**.
- On a **slope** the cart rolls **downhill** by itself; going up, it slows and may roll back. A Station Rail holds it still.
- An **empty cart** doesn't move on its own on the flat — but it rolls down any slope and carries on along a Boost Rail if it is already moving. Give it a push by riding it, then get out.
- At the **end of the line** the cart stops with a thud. If the rail under it is **broken** it drops off the track, and joins a rail again when it lands on one.
- The cart tilts on slopes and its wheels turn. Lava burns a cart up.
- **Hit it three times** to pick it up as a Minecart. Riders are saved with the world: reopen it and you are still on board.
- The **Distance by Minecart** statistic counts every metre.

Hills, and rails to craft



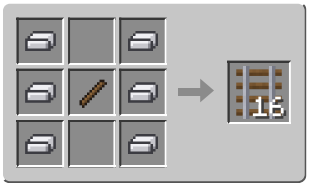
A sloped track: each rail is one block higher than the one before. The pieces tilt by themselves.



A Party Bag open. It holds nine kinds of things.

Tips for tracks

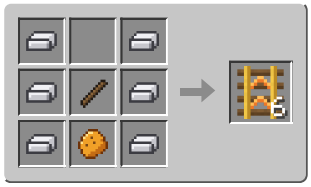
- A **circuit** with a Station Rail at one corner makes a little railway station: roll in, stop, press W, go round again.
- Put a **Boost Rail before a hill**: the cart arrives fast enough to climb it, and rolls back down on the other side.
- Leave a **flat piece** at the top and the bottom of every slope, and don't put a Boost Rail on a curve.
- You can ride through a tunnel, over water on a bridge, or round a tower — any ground that holds the rails.



Rail ×16

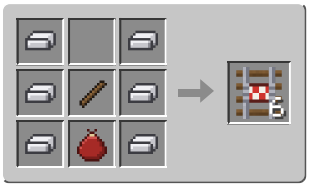
CRAFTING TABLE

6 iron and a stick make 16.



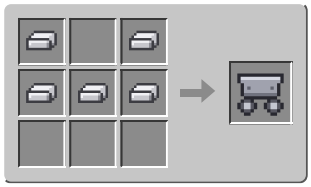
Boost Rail ×6

CRAFTING TABLE



Station Rail ×6

CRAFTING TABLE



Minecart

CRAFTING TABLE

Five iron ingots.

Party Bags

- A Party Bag is a **small bag with nine slots**: the things inside come with it wherever it goes, and are saved with the world. (A bag can't go inside a bag.)
- **Right-click** (tap) it to open it and fill it, taking things out the same way as a chest. Shift-click moves things in and out at once.
- **Sneak and right-click** to **tip it out** in front of you with a pop of confetti. The bag stays with you, empty.
- **Dye it** with 1–8 dyes in the crafting grid, just like leather armour, and wash the colour off by putting the bag in the crafting grid on its own. The contents stay inside.

Giving a Party Bag

- **To a villager:** hold the bag and right-click. The bag is theirs. The **food** inside goes into the village's food store (more children!), and your **standing** goes up — more the more kinds of things you put in. You earn **A Thoughtful Gift**.
- **To a Fellhound or a fox cub:** right-click it. It eats what it likes from the bag, a bite at a time (a wild one may be **tamed**), and you **keep the bag** with what it left. Good for a pet that needs feeding up.
- Confetti bursts either way.

Party things



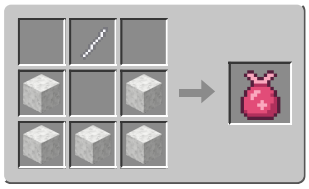
A party: a cake eaten down in stages, balloons, a string of bunting between two posts, and a Party Hat on an armour stand — in the middle of a confetti pop.



A villager is given a Party Bag: confetti, hearts and a thank you.

THING	HOW IT WORKS
Party Popper	Right-click (tap) in the air: a **bang and a burst of confetti**, and villagers within 9 blocks cheer. One is used up each time. Earns **Party Time!**
Party Cake	Put it on solid ground. Right-click a slice to eat it: **6 slices**, each filling you up by a drumstick; the candle glows until the cake is half gone. If you're full you can't eat any more. Earns **Let Them Eat Cake**.
Bunting	Hangs **under a solid block**; the string follows the way you were facing. Line up several for a string of flags.
Balloons	A bunch of red, yellow and blue balloons that stands on the ground. Break them with a hand or a tool.
Party Hat	Wear it on your head (it is armour with no armour points) or put it on an **armour stand**. **Dye it** any colour.

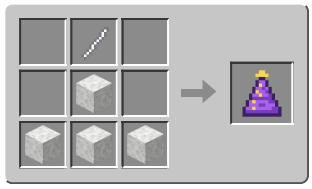
- All of these are **broken quickly by hand** and need something to stand on or hang from: take the support away and they pop off.
- Party Bags and Party Hats go with the leather armour recipes for dyeing: one dye to colour them, many to mix.



Party Bag

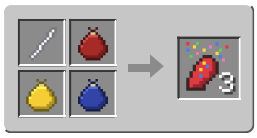
CRAFTING TABLE

Any wool (any colour mix).



Party Hat

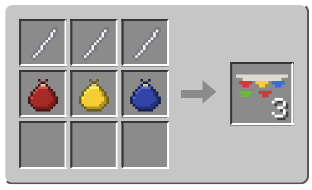
CRAFTING TABLE



Party Popper ×3

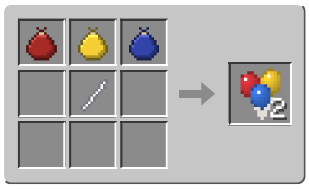
INVENTORY OR TABLE

Shapeless: any order.



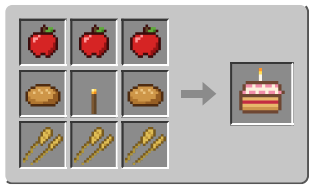
Bunting ×3

CRAFTING TABLE



Balloons ×2

CRAFTING TABLE



Party Cake

CRAFTING TABLE

Three apples, two bread, a torch, three wheat.

A party checklist

1. Make and fill a Party Bag (apples, bread, a cake…). 2. Put up bunting and balloons round the village well. 3. Set down a cake. 4. Pop a Party Popper — the villagers cheer. 5. Give the bag to a villager. 6. Share the cake, and put on the hat.

26 Village life

Every villager has a name and a life of their own. Villages remember how you treat them, grow with children when they have food and beds, meet round their bell at midday, and carry on while you are away.



Midday: the village gathers round the bell on the well. Look at a villager to see who they are.



Children playing by the well. They grow up in 20 minutes and then look for a job.

Names and the info card

- Every villager has a name, like **Marisa the Smith**. Villages have names too.
- Look at a villager (or the Sentinel) within 8 blocks: a card at the top of the screen shows the name, level and job, what they are doing, and the village with **your standing** there.
- The trade screen shows their name, and if a villager dies near you, you are told who it was.

Your standing

STANDING	POINTS	WHAT IT MEANS
Honoured	60 to 100	Prices up to 30% lower; villagers wave
Friendly	25 to 59	Prices a little lower; villagers wave
Neutral	−24 to 24	Normal prices (where you start)
Distrustful	−59 to −25	No trading; villagers keep away from you
Hostile	−100 to −60	Prices up to 30% higher, and the Sentinel attacks you

Raising it (and losing it)

A trade	**+1**	at most 12 a day
A gift	**+3**	at most 15 a day
A monster killed in the village	**+1**	at most 6 a day
A night raid defended	**+40**	
Hitting a villager (a child)	**−8**	(−15)
Killing a villager	**−30**	
Hitting / killing the Sentinel	**−5**	/ −20

Each dawn your standing drifts back towards neutral: 2 points a day from above, 4 from below.

> **Gifts**
>
> Sneak (**Shift**) and right-click a villager while holding **Bread**, an apple, a carrot, wheat, cooked meat or fish, or a flower. Children and villagers without a job take a gift with a plain right-click. Food goes into the village's food store.

Children and the food store

- Farmers' harvests and your gifts of food go into the village's **food store**.
- By day, while you are near: when the store holds **12 food**, there is a **free bed** and two grown-ups, a child is born (with hearts). You earn **A Growing Village**.
- Children are half size, have no job and can't trade. They play round the well and run home when frightened.
- After **20 minutes** they grow up and take a free workstation nearby.
- More beds means more villagers: build a house with a bed in the village.

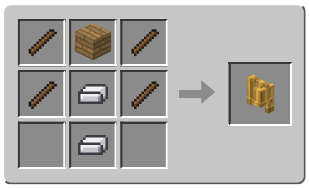
Habits

Villagers go indoors when it rains, and shut the doors of their houses at night.



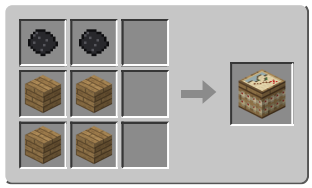
Ringing the bell at dusk: monsters lurking in the ground and behind the houses glow, seen through walls.

The Village Bell



Village Bell

CRAFTING TABLE



Map Table

CRAFTING TABLE

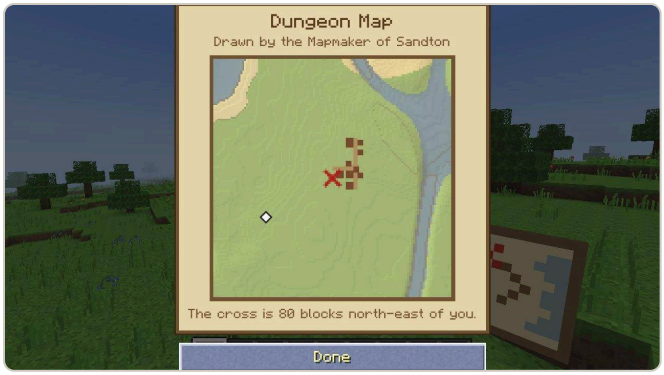
Mapmaker

- New villages (worlds made with 1.8) have one in the middle of the well's roof. Craft one to give any village a bell.
- **Midday:** the villagers gather round it (or round the well) to chat.
- **Ring it** (right-click): everyone hurries indoors for a minute, and monsters within 32 blocks **glow** for 30 seconds — you can see them through walls.

While you are away

When you come back to a village — or sleep through the night in it — it catches up on the time that passed: crops grow and the farmers harvest them, traders restock, newcomers move in, and children grow up or are born. A message tells you what happened.

The Mapmaker and explorer maps



A Dungeon Map: the land round the red cross, with rivers, villages and you (the white diamond).



Holding a map: the compass at the top left shows which way and how far.

- A **Map Table** gives a villager the **Mapmaker** job (new villages have one).
- Mapmakers sell a **Ruin Map** from the start, a **Dungeon Map** at Apprentice and a **Village Map** at Journeyman, and buy leather, feathers or string.
- Each map leads to a real place: the nearest one that village hasn't mapped yet, so the next map shows somewhere new. Dungeons only exist in worlds made with 1.7 or later.

- **Hold** a map: the compass shows how far and which way — near a dungeon, also how far down to dig.
- **Right-click** to look at the map itself.
- Reach the spot and the map is marked **found**: you earn **X Marks the Spot**.
- A map from the Creative catalogue is drawn the first time you use it, to the nearest place of its kind.

27 Dropbox sync

Play the same world on your Mac and on your iPad. Blockfell keeps every world in your Dropbox and brings each device up to date when you open a world, so you carry on exactly where you left off — on either device.



The world list shows where each world stands with Dropbox.



Singleplayer › **Dropbox…**: linked, with Sync Now and Unlink.

Set up once (about 5 minutes)

1. On a computer, go to **dropbox.com/developers/apps**, sign in and click **Create app**.
2. Choose **Scoped access** and **App folder**. Give it a name, e.g. *Blockfell Avizent* (names are unique on Dropbox). Click **Create app**.
3. **Permissions** tab: tick **files.content.write** and **files.content.read**, then **Submit**.
4. **Settings** tab: under **Redirect URIs** add **https://avizent.github.io/blockfell/**. Leave *Allow public clients (PKCE)* on **Allow**. Set the **App folder name** to **Blockfell**.
5. Note the **App key** (letters and numbers). Blockfell asks for it once per device, unless it is built in.

Set the permissions before linking. If you linked first, Unlink and link again.

Link each device

1. Open Blockfell from **https://avizent.github.io/blockfell/** — on the iPad, from its Home Screen icon. (Sync doesn't work in the Blockfell.html file or the claude.ai page.)
2. **Singleplayer › Dropbox…** (or Options › Dropbox Sync…). Paste the App key if asked.
3. **Link to Dropbox**: sign in to Dropbox, choose **Allow**, and you are back in Blockfell, linked.
4. Home Screen app: if Dropbox doesn't bring you back, use **Link With a Code…** — choose Allow, copy the code Dropbox shows and paste it in.

Link the **same Dropbox account** on every device. Worlds already on each device go up to Dropbox, and each device gets the others'.

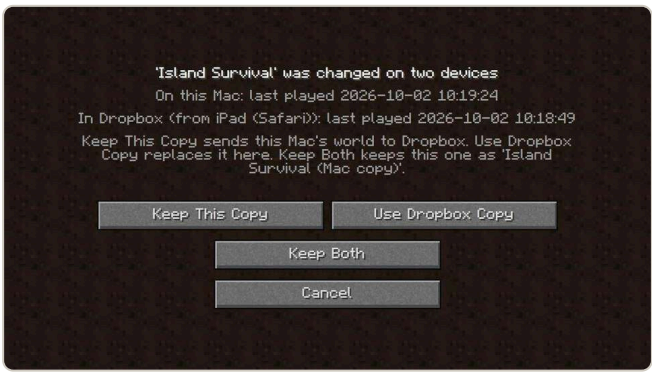
> **Your privacy**
>
> Blockfell can only see its own folder, **Dropbox › Apps › Blockfell**, never the rest of your Dropbox. **Unlink** removes the link from a device; Dropbox › Settings › Connected apps removes it everywhere.

How it keeps in step

- **Opening a world** (Play or Continue) first asks Dropbox for the latest copy. If another device saved it since, it is downloaded first: “Updated from Dropbox”.
- **While you play**, the world goes to Dropbox every 2 minutes, at once when you switch away (or the iPad sleeps), and on **Save and Quit**.
- **The world list** syncs everything when it opens: worlds from your other devices appear.
- **No connection?** Play as usual; the world goes up when you are back online.

What the world list says

In Dropbox, synced 14:32	Up to date.
Waiting to upload to Dropbox	Played here, not sent yet (offline).
Not in Dropbox yet	New: it goes at the next sync.
Changed on two devices: pick one	Choose a copy (right).
Only here (deleted from Dropbox)	Deleted from Dropbox on another device; this copy stays.
Dropbox copy needs Blockfell 1.x	Saved by a newer Blockfell: update this device first (Restart).



Played on both devices without a sync in between: you choose.

Changed on two devices

If a world was played on both devices without syncing in between (one was offline, say), Blockfell never overwrites either copy. Opening it asks: **Keep This Copy** (it goes to Dropbox), **Use Dropbox Copy** (it replaces this one) or **Keep Both** (this one becomes ‘*name* (Mac copy)’). Tip: Save and Quit before you switch devices.

Deleting

Delete Here Only keeps the Dropbox copy (Dropbox… lists it with a **Download** button). **Delete Everywhere** removes it from Dropbox too; Dropbox keeps deleted files for a while (dropbox.com › Deleted files).

28 Menus, saving and settings

Pause menu (Esc)

Back to Game	Carry on playing.
Advancements	Goals reached (listed below).
Statistics	Time played, blocks mined, distances, deaths and more.
Options...	Settings, plus difficulty and game mode for this world.
Sound: ON / OFF	All sound on or off, in one tap (or M).
Save and Quit to Title	Saves everything and returns to the title screen.

> **Saving**
>
> The game also saves every minute: your position, inventory, every block you changed, chests and creatures. Use **Save and Quit** before closing the tab: it also updates your Dropbox backup.

Options

FOV	How wide you see.
Render Distance	How far the world is drawn. Lower it if the game is slow.
Render Scale	A lower resolution, for speed.
Sensitivity, Invert Mouse	Mouse look speed and direction.
Brightness, GUI Scale	Moody to Bright; size of menus and display.
View Bobbing, Clouds	Turn head-bob and clouds on or off.
Sound, Music, Volume	All sound, or just the music, on or off; master, music and effects levels.
Reopen Last World	Closed mid-game? Start in that world.
Controls...	A reminder of every key.

Advancements (88 to collect)

- **Timber!** · Chop a log from a tree
- **Woodworker** · Turn a log into planks
- **Workbench** · Craft a crafting table
- **Pick It Up** · Craft a wooden pickaxe
- **Rock Solid** · Mine cobblestone with a pickaxe
- **Upgrade** · Craft a stone pickaxe
- **Hot Stuff** · Craft a furnace
- **Metal Age** · Smelt an iron ingot
- **Iron Will** · Craft an iron pickaxe
- **Suit Up** · Wear a piece of armour
- **Let There Be Light** · Place a torch
- **Armed** · Craft a sword
- **Monster Slayer** · Defeat a hostile creature
- **Sharpshooter** · Hit a creature with an arrow
- **Snack Time** · Eat something
- **Hoarder** · Open a chest
- **Deep Dive** · Go below height 20
- **Summit** · Climb above height 105
- **First Night** · Live to see the sun rise again
- **Sweet Dreams** · Sleep in a bed
- **Rune Carver** · Inscribe runes at a Rune Table
- **Green Fingers** · Harvest a ripe crop
- **Good Deal** · Trade with a villager
- **Hero of the Village** · Defend a village from a night raid
- **Storm Watcher** · See lightning strike close by
- **Loyal Companion** · Tame a Fellhound
- **A Splash of Colour** · Craft a dye or dyed wool
- **Signpost** · Write something on a sign
- **Art Lover** · Hang a painting
- **Set Sail** · Row a boat across the water
- **Catch of the Day** · Catch a fish
- **Cooling Off** · Pour water on lava to make Cinderstone
- **Dungeon Delver** · Open a chest in a dungeon
- **Cage Breaker** · Break a Monster Cage
- **A Growing Village** · See a child born in a village
- **X Marks the Spot** · Follow a map to the place it shows
- **Into the Cinderdeep** · Go through a Deepgate
- **Embers in the Dark** · Mine Ember Ore
- **Shrine Raider** · Open a chest in an Ember Shrine
- **Fireproof** · Make a Cinder Charm
- **On Display** · Put armour on an Armour Stand
- **Dressed to Impress** · Dye a piece of leather armour
- **Ember Piglet** · Breed two Ashboars with Glowcaps
- **Among the Stars** · Go through a Stargate to the Starhollow
- **Silence the Bells** · Ring all four Storm Bells
- **Free the Stars** · Defeat the Hollowdrake
- **Take Wing** · Glide on Starwings
- **Snowball Fight!** · Throw a snowball
- **A Frosty Friend** · Build a John
- **Downhill Dash** · Ride a Sledgy fast down a snowy hill
- **Best Friends** · Tame a Tim
- **What's Your Name?** · Give a creature a name with a Name Tag
- **Little Ones** · Feed two farm animals so they have a baby
- **Woolly Rainbow** · Dye a sheep
- **All Aboard** · Ride a Minecart along a rail
- **Express Service** · Whizz along on a Boost Rail in a Minecart
- **Party Time!** · Pop a Party Popper
- **A Thoughtful Gift** · Give a Party Bag to a villager or a pet
- **Let Them Eat Cake** · Eat a slice of Party Cake
- **Fungus Among Us** · Step onto a Spore Isle
- **Cap in Hand** · Shear a Sporecap
- **Glow Soup** · Eat a Spore Stew
- **Three Prongs** · Hit a creature with a thrown Trident
- **Saddle Up** · Climb onto a saddled pig, horse or Lavastrider

 **Lava Walker** · Ride a Lavastrider across lava
 Gallop! · Ride a horse 300 blocks
 **Long Reach** · Hit a creature with a spear from 4 blocks away
 **Loaded** · Fire a crossbow

 **Fire Arrow** · Set a creature alight with a Flame Arrow
 **Bubble, Bubble** · Craft a Brewing Stand
 **Down the Hatch** · Drink a potion
 **Splash Zone** · Throw a splash potion

29 Backing up your worlds

Keep copies of your worlds in Dropbox (or any folder you like). Each backup is a single **.blockfell** file holding everything needed to restore that world — usually a few KB to a few hundred KB.



The world list with **Export World**, **Import World…** and the backup reminder.



Import World lists every backup in your backup folder.

Set up once

1. Click **Singleplayer**, select a world and click **Export World**.
2. A folder window opens. Go to **Dropbox › 001 0125Vibe Coding › MineCraft** and click **Select**.
3. If the browser asks whether Blockfell may save changes to that folder, choose **Allow** (or **Allow on every visit** if offered).
4. The file **World Name.blockfell** appears in the folder and Dropbox copies it to the cloud. Blockfell remembers the folder from now on.

Automatic backups

- With **Auto Backup: ON** (the default), every **Save and Quit** updates that world's backup file. A "World backed up" message confirms it.
- Each world has its own file, replaced by each newer copy. Dropbox's version history can bring back an older one.
- Closing the tab without Save and Quit keeps your world in the browser but doesn't update the backup.
- The browser may ask for permission again at the start of a new session; click **Allow**.
- Switch automatic backups on or off on the Import World screen.
- Below the world list you see when the selected world was **last backed up**; when a backup is overdue, a yellow reminder suggests **Export World**.

Restoring or moving a world

1. Click **Singleplayer**, then **Import World…**. Your backups are listed with the world name, backup date, game mode and size.
2. Select one and click **Import Selected** (or double-click it).
3. If that world is already on this computer you can **Replace with Backup**, **Keep Both** (the backup is added as "… (imported)") or **Cancel**. Blockfell warns you if the copy on this computer has been played since the backup was made.

On another computer: open Blockfell.html there, click **Import World…** → **Choose Folder…**, pick the same Dropbox folder and import. **Import From File…** opens any single .blockfell file, for example one a friend sends you.

Which browser?

Chrome, Edge	Everything above: remembered folder, backup list and automatic backups.
Firefox, Safari	**Export World** downloads the file to your Downloads folder — move it into Dropbox › MineCraft. Import with **Import From File…**. No automatic backups.

> **1.10** updates your worlds to a new save format and keeps the old copy (**Edit › Restore Pre-1.10 Copy**). Update every device the same day.

30 Tips and troubleshooting

Tips

- Turn on **Bonus Chest** for your first world: it saves the slow first few minutes.
- Carry a stack of dirt or cobblestone. Pillar up by jumping and placing a block beneath you; block off creatures in an emergency.
- Explore caves with a stone pickaxe, a sword, food and plenty of torches. Place torches on one side only, so you can find your way back.
- Cook meat before eating it: cooked porkchop and steak fill more than twice as much as raw.
- Iron armour takes 24 ingots for a full set and cuts damage a lot. A chestplate first gives the most protection per ingot.
- Middle-click a block to swap it into your hand quickly while building.
- Use seeds to share worlds: a friend who types the same seed gets the same landscape.
- Place a bed early and sleep in it once: dying then brings you back home instead of to the world spawn.
- Carry a water bucket when exploring: pour it over a cliff edge and ride the waterfall down safely, then scoop the source back up.
- Carry a few pieces of raw meat: a Fellhound met on the way can become a companion for life.
- Plant a flower bed by your house: flowers are dyes, and bone meal on grass makes more of all four kinds.
- Put a sign at every cave entrance you explore, and a lantern at every junction, so you always find your way out.
- Travel by boat along rivers and coasts: it's faster than sprinting and costs no food. Carry the boat on with you when you land.
- Short of food? A fishing rod needs only sticks and string, and cooked fish are as good as most meat. Fish bite sooner in the rain.
- Going deep? Take a water bucket: it puts you out if you catch fire, and pouring it on lava in your way turns it to rock.

Troubleshooting

It won't open on iPhone or iPad	Open Blockfell from a web address, not the file (section 2).
The mouse doesn't turn my view	Click once on the game. If a menu is open, press **Esc** or **Back to Game** first.
The game is slow or jerky	In Options, lower **Render Distance** (try 6) and **Render Scale**; turn Clouds off.
My world has gone	Worlds live in the browser that made them. If its data was cleared, use **Import World…** with your backup.
A block won't drop anything	Stone-type blocks need a pickaxe, iron ore needs a stone pickaxe, and glass never drops.
I can't sleep	Only at night or in a thunderstorm, within 4 blocks of the bed, with no hostile creature close by.
My painting came out too big	Hit it and hang it again while sneaking (**Shift**) for a small one; right-click it for other pictures.
No Skybells or Moon Daisies	They grow wild only in worlds created with 1.4. In older worlds use bone meal on grass: all four flowers sprout.
My Fellhound has gone	It may still be sitting where you left it: go back and right-click it. Hounds far from you wait where they are until you return.
I can't sprint	Your hunger is too low (3 drumsticks or fewer). Eat something.
A villager won't trade	Your standing in that village is too low. Give gifts (sneak and right-click with food) and wait: it recovers a little each day.
I'm on fire	Jump into water or stand in the rain. It burns out after 15 seconds.
"Blockfell 1.x is ready"	A new version is ready: click **Restart**. Worlds are kept.
No sound	Check **Sound** is ON (pause menu, Options or **M**), then click in the game once: browsers start audio only after a click.
Checking performance	**F3** shows the frame rate; **#bench** after the address runs a benchmark.

31 Potions and the Brewing Stand

Bottle some water, add an ingredient and wait: a Brewing Stand turns plain water into a potion with a power of its own. Drink it yourself, or add Glimmer Dust and throw it.



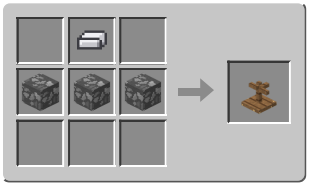
A Brewing Stand on solid ground.



Three bottles at the bottom, an ingredient at the top right and a piece of Coal for fuel at the top left. The green bar shows the brew.

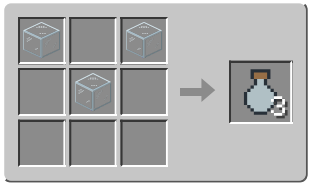
Making a potion

1. Craft **Glass Bottles** (three glass make three) and a **Brewing Stand** (an iron ingot over three cobblestone).
2. Hold a Glass Bottle and **right-click (tap) water** to fill it: a Water Bottle.
3. Place the stand and open it. Put up to **three Water Bottles** in the bottom slots, one **ingredient** on the right and a piece of **Coal** for fuel on the left. One Coal is enough for 10 brews.
4. Wait about **20 seconds**. The ingredient is used up and every bottle becomes the potion.
5. **Hold right-click** (press and hold on a tablet) to drink it.



Brewing Stand

CRAFTING TABLE



Glass Bottle ×3

CRAFTING TABLE

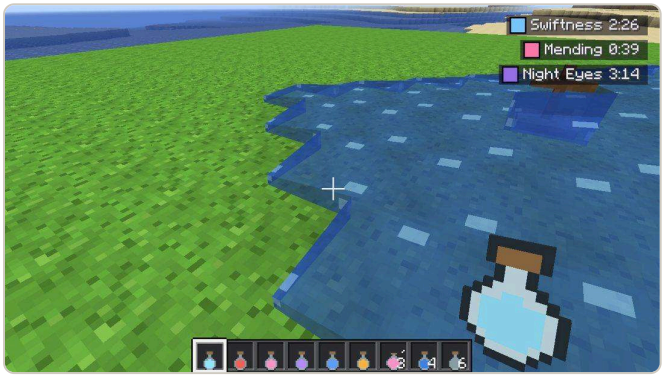
Good to know

- The effects you have are listed at the **top right** of the screen with the time left. Drinking the same kind again never adds up: it lasts as long as the longer of the two.
- Effects are saved with your world, and you lose them if you die.
- **Night Eyes** brightens caves and the night. **Emberward** makes lava harmless, but you can still fall. **Featherfall** stops fall damage altogether. **Deep Breath** stops your air running out.

The seven potions

POTION	INGREDIENT	LASTS	WHAT IT DOES
Potion of Swiftness	**Feather**	3 min	Run a third faster.
Potion of Might	**Bone**	3 min	Your blows hit 1.5 hearts harder.
Potion of Mending	**Starbloom**	45 s	Heal half a heart every two seconds.
Potion of Night Eyes	**Glowcap**	4 min	See in the dark.
Potion of Deep Breath	**Drift Silk**	4 min	Breathe under water without running out of air.
Potion of Emberward	**Ember**	4 min	Fire and lava cannot hurt you.
Potion of Featherfall	**Gloomdust**	4 min	Fall from any height and take no damage.

Splash potions



Three effects running at once, listed at the top right.



A splash potion bursts into coloured sparkles.

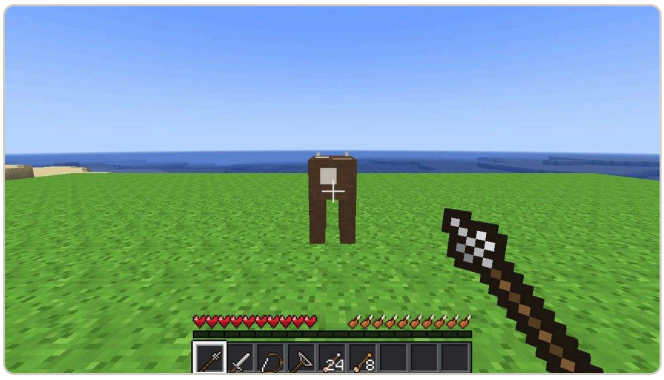
- Put a finished potion in the stand with **Glimmer Dust** as the ingredient to make a **Splash Potion**.
- **Right-click** to throw it. It bursts where it lands and affects **you** if you are within 4 blocks — fully in the middle, and for three quarters as long at the edge.
- A splash of **Mending** also heals creatures nearby (3 hearts), **including hostile ones**, so keep it for friends. A splash of **Emberward** puts out burning creatures. The other splashes only affect you.
- Breaking a stand spills what is inside it.

Advancements

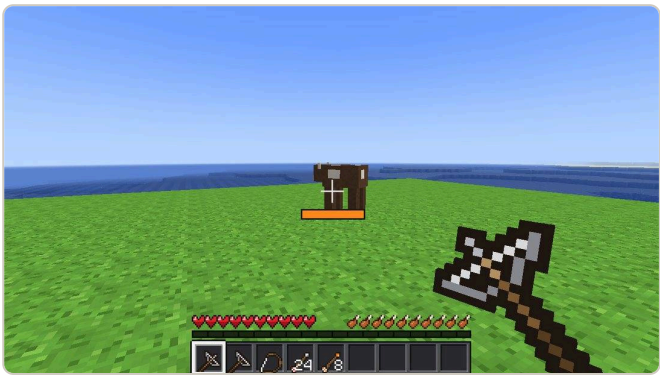
- Three to collect: **Bubble, Bubble**, **Down the Hatch** and **Splash Zone**.

32 Spears, crossbows and fire arrows

Three new ways to fight: a spear that reaches further than a sword, a crossbow that hits harder than a bow, and arrows that set things alight.



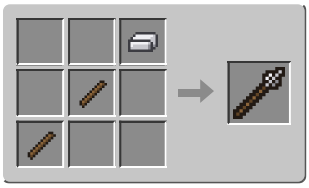
An Iron Spear reaches a cow four blocks away — a sword would fall short.



A loaded Crossbow: the bar under the cross is full and orange.

The Spear

- Made from **one material and two sticks** set on a slant: planks, cobblestone or iron.
- **Reach 4.5 blocks** (a sword reaches 3). Damage is a little lower: wood 3.5, stone 4.5, iron 5.5.
- It is **slower**: about two thirds of a second between thrusts, and a second click inside that time does nothing.
- **Run at them:** a thrust while sprinting does **2 extra**.
- Takes the same runes as a sword (Keen, Sturdy, Plunder). Hitting from 4 blocks earns **Long Reach**.

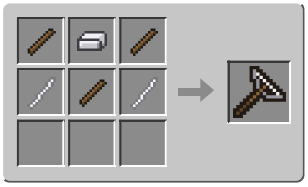


Iron Spear (or planks or cobblestone)

CRAFTING TABLE

The Crossbow

- Three sticks, an iron ingot and two string.
- **Hold right-click** (press and hold on a tablet) to **wind** it: about a second and a quarter, with a green bar under the cross. One arrow is used up and it becomes **loaded**.
- A loaded crossbow can be carried about. **Right-click** (tap) to fire: a fast, flat bolt that hits for about **10** (the bow's best is 8).
- It needs no arrows in **creative** mode. Earns **Loaded**.



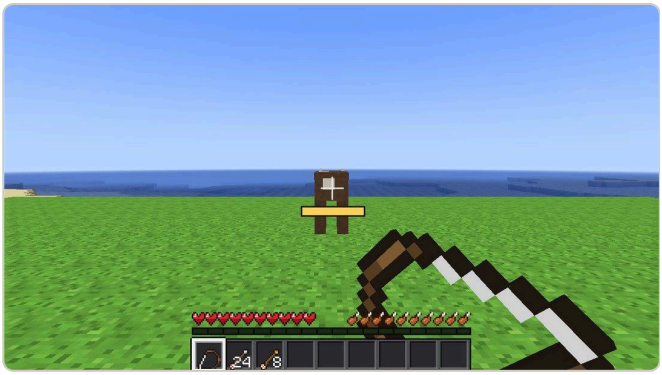
Crossbow

CRAFTING TABLE

The bow, and Flame Arrows

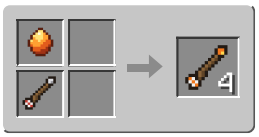


A Flame Arrow sets a creature alight; the arrow burns out where it lands.



Drawing a bow: the bar fills, and turns gold at full strength.

- The **charge bar** appears under the cross while you draw a bow or wind a crossbow. A bow is strongest when the bar is gold.
- A **Flame Arrow**: one **Ember** and four arrows make four. It sets what it hits **alight** (a few hearts of burning) and earns **Fire Arrow**. Water or rain puts the fire out.
- A Flame Arrow **burns out** where it lands, so you can't pick it up again.
- **Which arrow is used?** A Flame Arrow on your **hotbar** goes first; otherwise a plain Arrow; otherwise a Flame Arrow from the pack. Keep your Flame Arrows in the pack and move a few to the hotbar when you want them.
- Works with the bow and the crossbow, and with the Might rune on either.



Flame Arrow ×4

INVENTORY OR TABLE

One Ember and four Arrows, anywhere in the grid.

Which to choose?

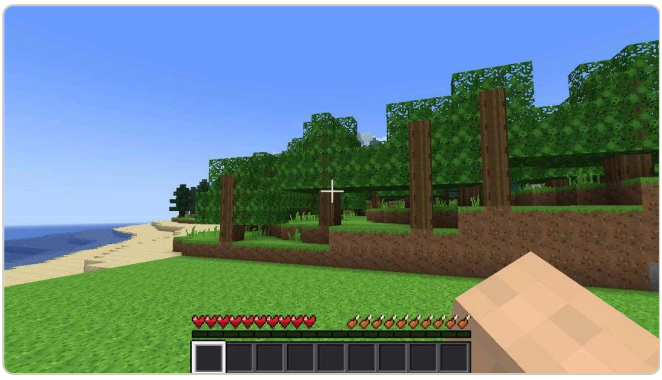
WEAPON	GOOD FOR
Iron Sword	Quick blows, the best damage up close.
Iron Spear	Keeping fast or leaping creatures at a distance; running thrusts.
Bow	Many arrows, quick to use, fine at long range.
Crossbow	One powerful flat shot, loaded in advance and kept ready.
Flame Arrow	Creatures that stay still; remember they burn too, so keep clear of dry places.

33 Saddles, horses and the Lavastrider

Put a saddle on a creature and ride it. Pigs, horses and the fire-loving Lavastrider all carry a rider, and the Lavastrider can stride across lava.



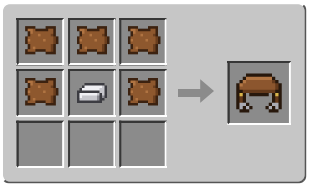
A saddled pig and a saddled horse. The saddle is visible on the creature's back.



Riding a horse: you look where you want to go, and it follows.

Saddling and riding

- Craft a **Saddle**: five leather and an iron ingot. **Right-click** (tap) a grown-up pig, horse or Lavastrider holding it. The saddle is used up and stays on, even after you save and quit.
- **Right-click it again** with an empty hand to climb on. Earns **Saddle Up**. A piglet can't be saddled.
- **Steer by looking:** hold **W** (or push the stick) and the creature turns to face where you look. **S** slows it. **Space** (the Jump button) makes it hop.
- To get off press **Shift** (the Sneak button on a tablet). To take the saddle off, **sneak and right-click** it with an empty hand: the saddle drops.
- If the creature is hurt badly enough to die, you are put back on your feet.



Saddle

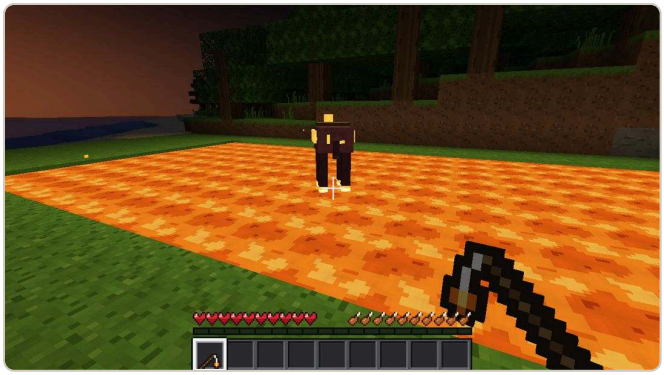
CRAFTING TABLE

Which to ride?

CREATURE	HOW IT GOES
Pig	A comfortable trot, about 4 blocks a second. Steps up one block and hops.
Horse	Much faster, about 7 blocks a second, and a high jump that clears a fence. It comes in five coats and lives on grassy plains. **Gallop!** is for 300 blocks ridden.
Lavastrider	Slow on land, about 3 blocks a second, but about 5 across lava. Needs an **Ember Lure** in your hand to steer.

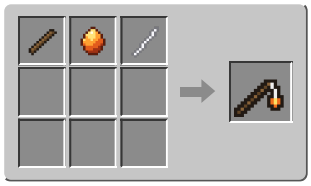
The game counts **Distance Ridden** and **Distance by Horse** in your statistics.

The Lavastrider and the Ember Lure



A saddled Lavastrider standing on the lava, glowing along its seams.

- Lavastriders wander the **lava seas** deep underground, a few at a time. They are fireproof and walk on the surface of lava as if it were a floor.
- A Lavastrider does not obey a rider by itself. Hold an **Ember Lure** (a stick, string and an **Ember**, in any order) and it follows where you look, as if chasing the glow. Put the lure away and it goes back to wandering.
- You are **not burnt** while you sit on it over lava. Crossing lava earns **Lava Walker**.
- **Careful:** if you climb off over lava, you land in the lava. Leave it at the shore.



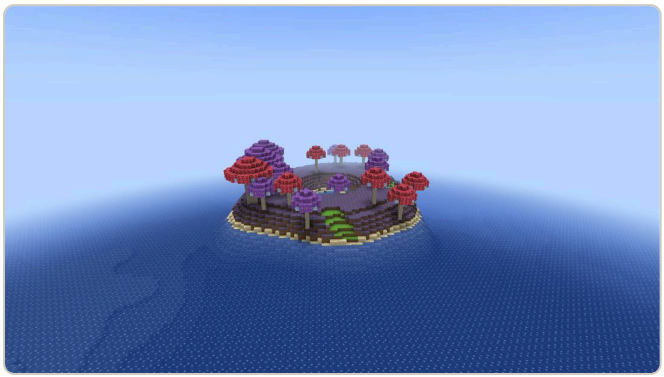
Ember Lure

CRAFTING TABLE

A stick, a string and an Ember, anywhere in the grid.

34 Spore Isle, Sporecaps and the Trident

Far out at sea, a few small islands of purple soil are covered in giant mushrooms. Gentle Sporecaps live there, and nothing hostile ever spawns. (The names are working names and may change.)



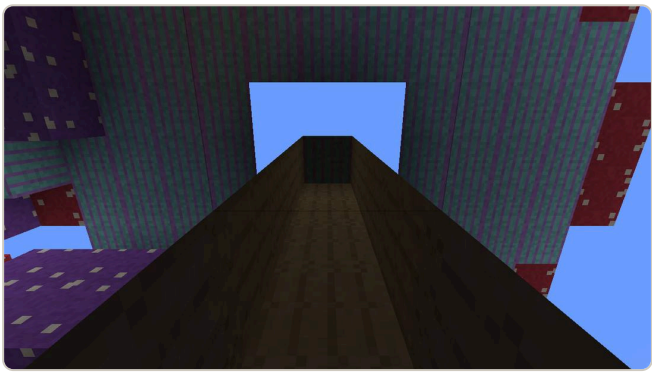
A Spore Isle seen from the air: a ring of land around a small lagoon.



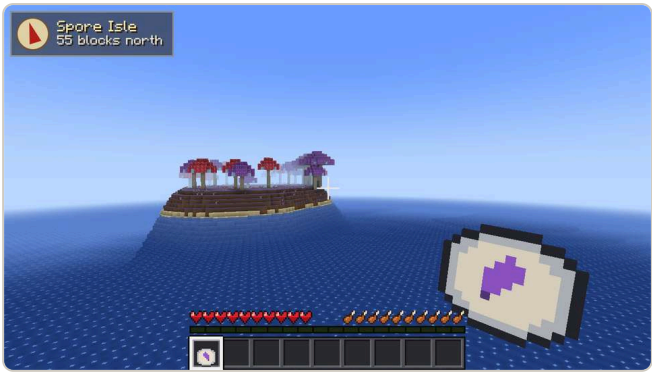
A giant mushroom cap, high above the island.

Finding and exploring it

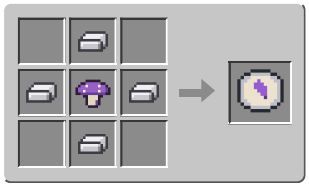
- A Spore Isle sits **in open ocean**, roughly one every thousand blocks or so. Sail out and look for bright purple and red tops over the water. Stepping onto one earns **Fungus Among Us**.
- The ground is **Sporeloam**, a dirt you dig with a shovel. Nothing grows from grass seeds here; glowing **Spore Tufts** sprout instead and give a Sporecap when broken.
- Mushrooms come in two sizes. Small ones are a stem and a cap. **Giant ones are hollow**: walk in through the doorway on the south side and look up at the glowing **Glowgills** under the cap.
- **Violet** and **Crimson** caps and **Spore Stems** can all be collected. Four stems make four planks.
- **Spore Compass:** four iron and a Sporecap. Hold it and a panel appears with an **arrow** and the distance to the nearest Spore Isle (up to about 3,500 blocks). Turn until the arrow points straight up and sail that way. On the isle it says so.
- **Only new worlds** have Spore Isles. A world you made earlier keeps its terrain exactly as it was.



Inside a giant mushroom, looking up through the stem.



The Spore Compass: Spore Isle, 55 blocks north.



Spore Compass

CRAFTING TABLE

Sporecaps

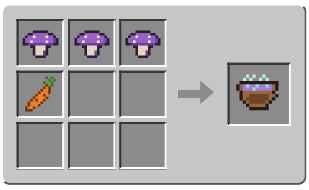


Two Sporecaps: one with its cap, one sheared bare.

- A small, peaceful mushroom creature. It lives only on Sporeloam, wanders slowly and is never afraid.
- **Shear it** with Shears to take the cap: you get 1 to 3 **Sporecaps** to eat or cook with. Earns **Cap in Hand**. The cap **grows back** after about three minutes. Shearing a bare Sporecap does nothing.
- Feed it a Sporecap and it falls in love. Two in love will make a baby, as with the other farm animals.
- A raw Sporecap is a small snack (2 hunger).

Spore Stew and the Spore Lamp

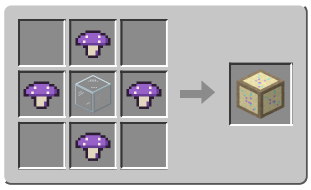
- **Spore Stew** fills 8 hunger and gives **Night Eyes** for 90 seconds, so you can see in the dark. Earns **Glow Soup**.
- The **Spore Lamp** is the brightest light there is (level 15) and looks like a glowing mushroom in glass.



Spore Stew

CRAFTING TABLE

Three Sporecaps and a carrot, anywhere in the grid.



Spore Lamp

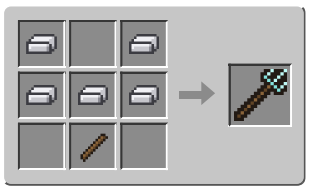
CRAFTING TABLE

The Trident



Drawing a Trident: the bar fills, and letting go throws it.

- In the hand it is a **spear** with long reach and a strong hit (6.5).
- **Hold right-click** (hold your finger on the screen) to draw it back and **let go** to throw. A quick tap does not throw. A full draw flies far and hits hard (7 to 8.5).
- It drops where it hits, **worn by one**, or sticks in a wall. Walk over it to pick it up. It lasts 300 uses. In creative it is never used up.
- Earns **Three Prongs** on the first hit.



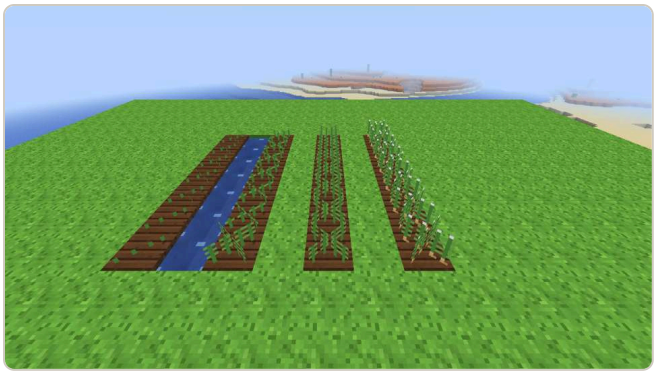
Trident

CRAFTING TABLE

35 Potatoes, Milk, night monsters and Sporecap colours

A new crop, a way to wash potion effects away, two new monsters of the night, and Sporecaps in three colours. (The monster names are working names and may change.)

Potatoes



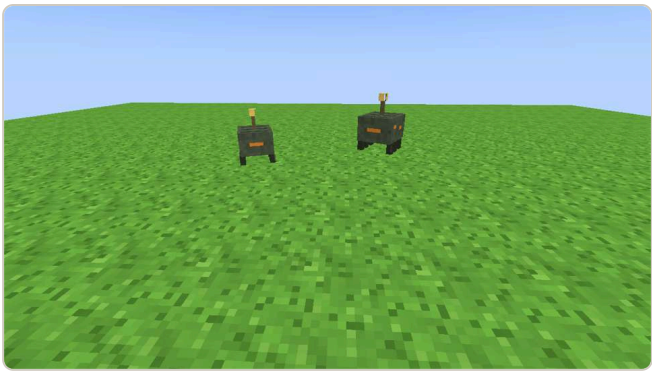
Potatoes in four stages, from a seedling to ready to dig.

- Plant a **Potato** on **farmland**, just like a carrot. It grows in four stages and is ready when the plant is tall with pale flowers. Digging a ripe crop gives **2 to 4 potatoes**.
- Raw, a potato fills 2 hunger. **Cook it in a furnace** and you get a **Baked Potato** that fills **5 hunger**.
- New potatoes only turn up in new farms you plant yourself: villages keep their wheat and carrots.

Milk

- Right-click a **grown-up cow** while holding an empty **Bucket**. You get a **Milk Bucket**. A calf cannot be milked.
- **Hold use to drink it.** The milk **clears every potion effect** at once (good or bad), and you get the empty bucket back. You cannot pour it out.

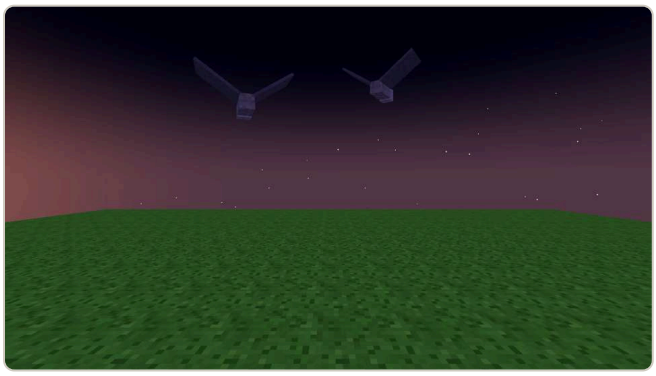
Powderbug



Two Powderbugs. The one on the right has lit its fuse and is swelling up.

- A round beetle with a fuse on its back. It hunts you at night (about one in eight dark-spawn monsters, never in deserts).
- It never bites. When it gets within about two blocks it **stops, hisses, swells and flashes** for 1.5 seconds, then **pops**.
- The pop hurts you and any creature close by (up to 7 hearts at point-blank range, less further away). It only breaks **soft natural blocks**: dirt, grass, sand, gravel and snow. **Planks, stone, doors, glass and chests are never harmed**, so a house is safe.
- **Run away** (about six blocks) and it fizzles out. **Kill it before the fuse runs out** and nothing happens. It drops **Blast Powder**.
- Blast Powder and string make 2 **Party Popper**.

Gloommoth



Two Gloommoths over the grass at night.

- A dark moth with glowing eyes. It flutters out of the **night sky** (never in the daytime, never on Spore Isle, never in peaceful mode). At most two at once.
- It **circles above you, swoops down for one bite** (3 damage on normal) and climbs away again. Watch the sky and hit it as it dives.
- At sunrise it flutters away. It drops **Gloomdust**.
- **Gloomdust brews a Potion of Featherfall** (no fall damage for 4 minutes). Glowcap still brews Night Eyes.

Sporecap colours



Violet, Crimson and Pale Sporecaps.

- **Violet** is the usual one (about 7 in 10). Shearing gives 1 to 3 Sporecaps; the cap grows back in 3 minutes.
- **Crimson** is uncommon (about 2 in 10). It gives **2 to 4** Sporecaps and its cap grows back in **2 minutes**.
- **Pale** is rare (about 1 in 12). Its cap **glows softly in the dark**. Shearing gives 1 to 2 Sporecaps and half the time a **Glowcap**.
- **Breeding:** a baby takes after one of its parents. About one baby in twelve is a surprise colour, so breeding two Violets can give a Pale.
- A sheared Sporecap now really loses its cap until it grows back. Sporecaps from older saves come back Violet.

Old worlds: all of this works in worlds you made before. Potatoes, milk, the monsters and the colours need no new world.

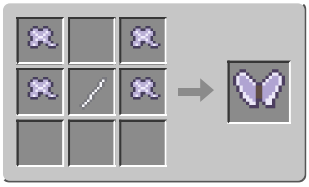
36 Wings, Fireworks, Fish Jars, Spring Pads and the Hootlet

An early glider, rockets for the night sky, a whole new bunch of fish with jars to keep them in, a bouncy pad, and an owl friend. (The Hootlet is a working name.)

Moth Wings



Moth Wings worn in the chest slot, with the silk and string that make them.



Moth Wings

CRAFTING TABLE

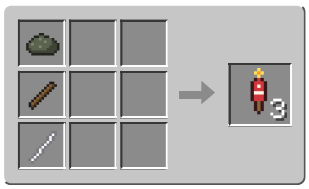
Four Moth Silk and a string.

- **Moth Silk** drops (0 to 2) from every **Gloommoth** you beat at night.
- Wear **Moth Wings** in the **chest slot**. **Jump in mid-air while falling** to open them and glide. You can start in the air only; you cannot take off from the ground.
- Look **down** to dive and pick up speed, **level out** to glide, look **up** to climb a little. A landing never hurts.
- They are the **light version of** **Starwings**: they fall a bit faster, so you travel less far for each block you drop, and they wear out after about **160 seconds** of gliding (Starwings last 432). They cannot be mended.
- Wings and Spring Pads are great together: bounce high, then glide.

Fireworks



Two rockets bursting over the grass at night.



Firework ×3

CRAFTING TABLE

Three of the ingredients in any order make 3 Fireworks.

- Right-click with a **Firework**. A rocket shoots straight up, trailing sparks, then bursts: a **ball**, a **ring**, a golden **willow**, a **two-colour ball** or a **heart**, in random colours.
- **Sneak and right-click** to launch a **volley of three** side by side (it uses up three).
- Harmless: they give only light and a bang. Night is best. In creative mode they are never used up.
- **Blast Powder** comes from **Powderbugs**.

Fishing and Fish Jars

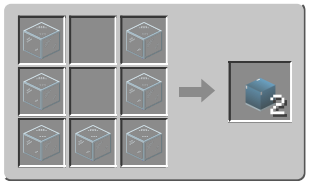


A jar for each kind of fish. The Glowfin, Moonfin and Sunfin jars give off light.

- The old Fishing Rod now brings up **five new fish** as well as Trout and Perch. **Where and when** you fish decides which:

FISH	WHERE YOU FIND IT	LIGHT IN A JAR
Raw Sunfin	Daytime, anywhere	5
Raw Moonfin	Night time, anywhere	7
Raw Frostfin	Snowy lands and taiga	3
Raw Glowfin	Spore Isle	12
Raw Ruby Carp	Rare, anywhere, any time	4

- Raw fish fill 2 hunger. **Cook them in a furnace** for **Grilled Fish** (6 hunger).
- Catching a Ruby Carp earns **Red Letter Day**.



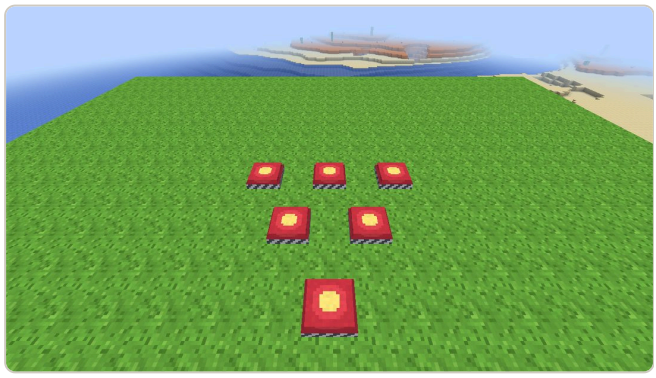
Fish Jar ×2

CRAFTING TABLE

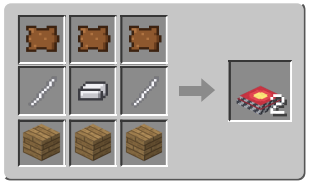
Five glass make 2 jars.

- Put a **Fish Jar** down like any block. **Right-click it holding a raw fish** to put the fish inside. Any of the seven fish will do (Trout and Perch too).
- **Right-click a full jar** to take the fish out again. Breaking a full jar gives you **both the jar and the fish**.
- Jars of Sunfin, Moonfin, Frostfin, Glowfin and Ruby Carp **give off light**: a Glowfin jar is a pretty lamp.

Spring Pad



Spring Pads laid out on the grass.



Spring Pad ×2

CRAFTING TABLE

Leather, string, iron and planks make 2 pads.

- Land on a **Spring Pad** and you **bounce back up**, and **never take fall damage**. The harder you land, the higher you go (up to about 16 blocks).
- **Sneak** to land softly with no bounce. It is only about 5/16 of a block high, so you can walk onto it.

Hootlet



A wild Hootlet (left) and a tamed one with its red ribbon (right).

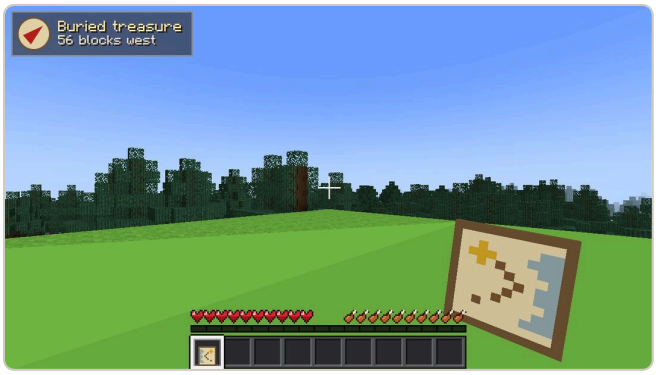
- A small owl that lives in **forests, birch woods and taiga**. Wild ones are **shy**: they fly off unless you **creep up (sneak)**.
- **Tame it:** right-click with **Wheat Seeds**. One try in three works; a try that fails uses up one seed. A tamed Hootlet has a **red ribbon** and 7 hearts, and the Hootlet spawn egg is in the creative menu.
- **Follows you**, fluttering round your head. **Stand still** for about three seconds and it settles on your **shoulder**.
- **Hoots a warning** (in chat, with sparkles and a call) when a hostile creature comes within 14 blocks of you. It does not warn again for 30 seconds.
- **Brings presents** now and then, usually a **Feather**, sometimes string, a bone or amber.
- **Right-click with an empty hand** to make it stay where it is; again to call it. Seeds mend it. A Name Tag gives it a name. It never wanders off, and it is saved with your world.
- A Hootlet is a lookout, not a fighter: it keeps out of the way. Arrows and snowballs pass it by.

Old worlds: all of this works in worlds you made before. Wings, fireworks, fish, jars, pads and Hootlets need no new world. Older fishing rods now bring the new fish too.

37 Treasure Maps, weather, music and the Wandering Trader

A reason to go exploring, a sky with moods, a way to write your own tunes, and a stranger who turns up with a great pack.

Buried treasure and Treasure Maps



A drawn Treasure Map in your hand: the compass in the corner points to the treasure and says how far.

- About **one chunk in fifty** of dry land hides a **buried chest**, four to six blocks under the grass. Nothing marks the spot, so you need a map.
- A **Treasure Map** is sold by the **Mapmaker** at the Expert level (**8 Amber**) and by the **Wandering Trader** (12 Amber). Now and then one turns up in a **ruin chest**.
- **Hold the map**: the corner compass shows **Buried treasure**, which way it is and how many blocks. It works the same way as the Spore Compass. **Right-click** to look at the map itself. **Dig down** where the compass shows a cross and the distance is zero.
- The chest holds **6 to 15 Amber**, iron, rune shards, food, arrows, bone meal and more. Sometimes a **rune-carved** iron tool or bow, a spare map, or a pair of **Moth Wings**.
- Never under rivers, the sea, villages or Spore Isles.

New worlds only: treasure is buried in worlds made in **2.11 or later**. Older worlds say so when you try to use a map. Make a new world to hunt for treasure.

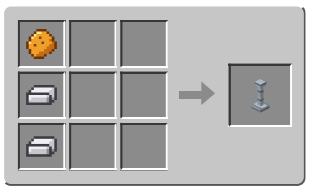
Weather



A storm bolt lands on a Lightning Rod and does no harm.

Rain, thunderstorms and lightning were already part of the game. 2.11 adds more to them:

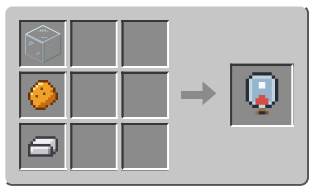
- **Wind** gusts and rushes in thunderstorms, so you can hear one coming.
- A **rainbow** can appear in the sky **opposite the sun** for a few minutes after rain stops in the daytime.
- **Lightning Rod**: set one on the **highest point** of a roof or tower. A bolt that was about to land **within 16 blocks** goes to the rod instead, in a shower of stars. Nothing is burnt or hurt. Make it with 1 Amber on top of 2 Iron Ingots. The rod must be the top block of its column.
- **Weather Glass**: hold it to read the weather in the corner: **clear**, **rain** or **storm** and about how many **minutes are left**. It tells you when the Weather Cycle rule is switched off.



Lightning Rod

CRAFTING TABLE

Amber on top, two Iron Ingots below.

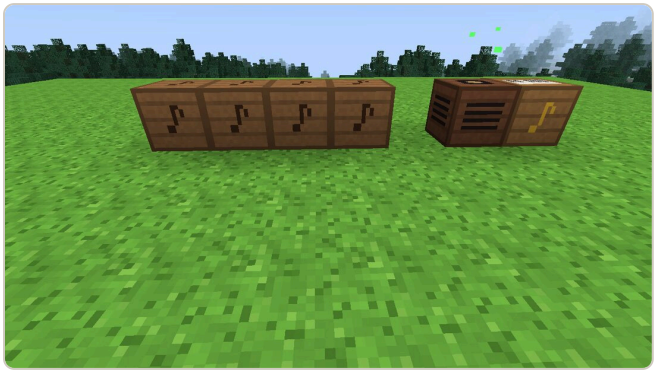


Weather Glass

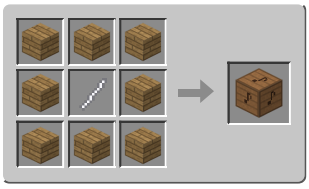
CRAFTING TABLE

Glass, Amber, Iron Ingot in a line.

Note Blocks, the Jukebox and the Tune Box

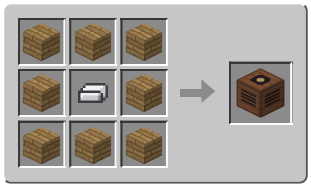


Note Blocks in a row, with a Jukebox and a Tune Box beside them.



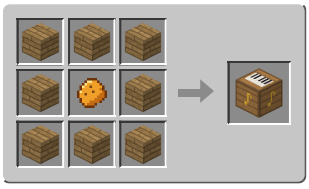
Note Block

CRAFTING TABLE



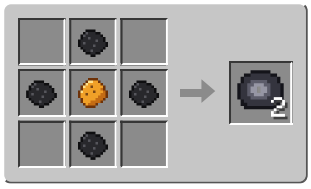
Jukebox

CRAFTING TABLE



Tune Box

CRAFTING TABLE



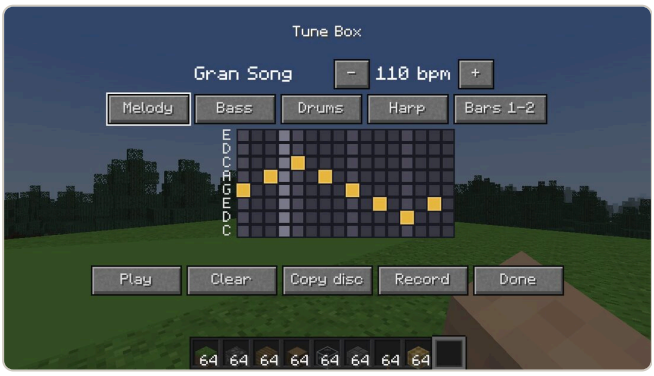
Blank Disc ×2

CRAFTING TABLE

Four Coal and an Amber make 2 Blank Discs.

- **Note Block**: **right-click** to raise its note one step and hear it (it goes through two octaves, from low C to high C, then starts again). **Punch** it to play it. The **block under it** picks the instrument: **wood** bass, **stone** thud, **grass** harp, **glass** bell, **wool or snow** pipe, **sand** snare drum, **gravel** hi-hat, anything else marimba.
- **Jukebox**: hold a **Music Disc** and **right-click** the Jukebox. The tune plays **round and round** and can be heard from some way off. Right-click it again to take the disc out. Breaking it gives the disc back.
- Three tunes are made by the game: **Fellfire Jig Disc** (a lively marimba jig), **Moss Lullaby Disc** (a slow pipe tune) and **Starhollow Drift Disc** (bells). The **Wandering Trader** sells them, and they are in the creative menu.

Write your own tune



The Tune Box: a grid of 16 squares across, two pages of them (32 steps). The bright column is where the music is now.

- **Right-click a Tune Box** to open it. Tap a square to put a note in it; tap again to take it out. Tap a **Melody** note and you hear it.
- **Melody** (8 notes, C to E), **Bass** (4 low notes) and **Drums** (kick, snare and hat) each have their own page. **Bars 1-2 / 3-4** switches between the first and second half.
- **- / +** change the speed (40 to 200 beats a minute). The **instrument button** chooses Harp, Bell, Pipe or Marimba for the melody. Type a **name** (up to 20 letters).
- **Play** plays your tune while you edit. **Clear** wipes the grid. **Copy disc** brings in the tune from a Music Disc in your hand, so you can change someone else's.
- **Record**: hold a **Blank Disc** and press Record. It becomes a **Music Disc** with your tune and its name. Put it in a Jukebox, give it away, or play it at the next family party.
- The tune and the Tune Box are **saved with the world**, and the disc keeps its tune when you move it.

The Wandering Trader



The Wandering Trader: an orange hood, a white beard and a big pack on his back.

- Now and then, **by day and never in a storm**, a Wandering Trader turns up **24 to 40 blocks** from you in the overworld. A line in the chat says so. Look for the **orange hood**.
- He stays about **nine minutes**, then packs up with a puff of smoke and walks on. **He never goes while you are trading.** Only one at a time, and a good while between visits.
- **Right-click** him to trade. He wants **Amber**.



His offers. He has no workstation, so a sold-out offer stays sold out.

- **Always:** **Potato** (8 for 1 Amber), **Milk Bucket** (2 Amber), and he **buys feathers** (12 for 1 Amber).
- **Four rare finds** picked at random each visit from: Treasure Map, the three Music Discs, Blank Discs, Fish Jars, Moth Silk, Fireworks, a Spore Compass, Sporecaps, Spring Pads, Rune Shards, a Name Tag, Ruby Carp, Lanterns, and he buys Glowfin.
- Trading with him earns the **Haggler** advancement.

Old worlds: the trader, weather, Note Blocks and the Tune Box all work in worlds you made before. Only buried treasure needs a **new world**.

Six new advancements (84 in all): Buried Treasure, Safe and Sound, First Note, Composer, Disc Jockey and Haggler.

38 Axolotls, Squid and Guardians

Three new creatures live in the water: a friendly pond salamander, a drifting squid, and a spiny guard of the deep sea that shoots a beam.



Axolotls in their five colours: pink, brown, gold, cyan and (very rarely) blue.

Axolotl

- A small smiling salamander with frilly gills. It paddles about in **lakes, rivers and the shallow sea** (water up to eight blocks deep), by day and night. Most are **pink, brown or gold**; **cyan** is uncommon and **blue** turns up about once in a hundred.
- **Hold a raw fish** (Raw Trout, Raw Perch or any other raw fish) while you are in or beside the water and nearby Axolotls swim over to you. **Right-click** one with the fish to feed it: it mends **4 health** and shows hearts. You earn **Pond Pal**.
- **It plays dead.** Hurt, an Axolotl may roll onto its side and lie still for twelve seconds, healing as it goes.
- **It fights for you.** An Axolotl bites any **hostile creature that is in the water** within nine blocks (2 damage a bite). If it finishes one off you get **Mending for 15 seconds** and the **Fighting Fit** advancement.
- Out of the water it flops about and, after about a minute, starts to dry out. It does not follow you on land. A **Name Tag** works on it.

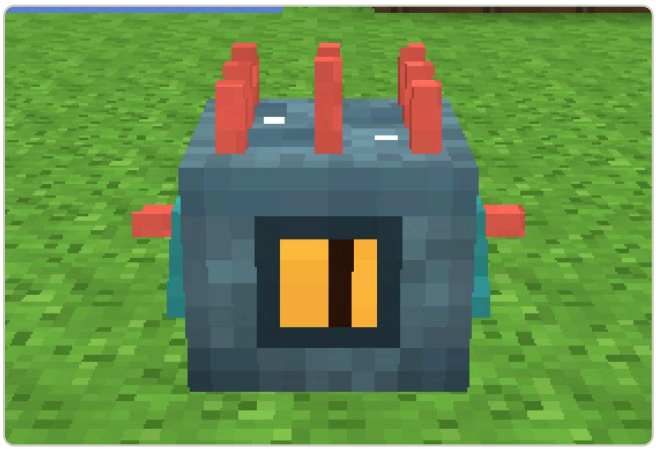
Squid



Squid: slate-blue, with big pale eyes and eight arms.

- Squid drift through **open water at least three blocks deep** and move in little pulses, squeezing and then gliding.
- **Hurt, a Squid blows a cloud of ink** and jets away from you for a few seconds. They are harmless and calm.
- A Squid drops **1 to 3** Ink Sac. One Ink Sac makes one Black Dye, in the crafting grid on its own. Picking one up earns **Inky Fingers**.
- Out of the water a Squid flops about and **suffocates in about six seconds**.

Guardian



The Guardian: a barnacled shell, a ring of coral-red spines and one amber eye.

- Guardians live in the **deep sea** (water at least seven blocks deep), well away from Spore Isle. They are hostile, so there are none in **Peaceful**, and they never leave the water. A Guardian has **30 health**.
- It keeps a distance of about seven blocks and **fixes its amber eye on you** when you are within **14 blocks** and it can see you, even if you are on the shore. A **beam** grows out of its eye over about **three and a half seconds** and then **zaps you**: **1.5 hearts** on Easy, **2.5** on Normal, **about 3.75** on Hard. It then needs two seconds before it can start again.
- **Break its line of sight** (swim behind rock, or put a block between you) and the beam charge starts again from nothing.
- **Its spines prick:** hit one when you are within about three blocks and you lose **1 heart**. A bow, a crossbow or a trident from further off is safer.

What a Guardian leaves

- 0 to 2 **Tide Shard**, a raw Trout now and then, and 10 experience. Defeating one earns **Deep Sea Duty**.
- **Tide Shards** brew a **Potion of Deep Breath** (Deep Breath: breathe underwater) in the Brewing Stand, in place of Drift Silk.
- The **Wandering Trader** sometimes buys **Tide Shards** (2 for 4 Amber) and **Ink Sacs** (6 for 2 Amber).

Tips

- Take a Potion of Deep Breath, or fight from a boat or the shore.
- Bring an **Axolotl**: it bites hostile things in the water, though a Guardian is a hard fight for one.
- All three have **Spawn Eggs** in the creative menu (**Axolotl Spawn Egg** **Squid Spawn Egg** **Guardian Spawn Egg**). Names, colours and shapes are our own.

Old worlds: they all appear in worlds you made before. Also fixed in 2.13: a **tamed Hootlet** is now remembered when you save and reload a world.

Four new advancements (88 in all): Pond Pal, Fighting Fit, Inky Fingers and Deep Sea Duty.